

# YFG FENDER HOMOLOGATION & BOP CHECKING MACHINE

CV-X300 Series

## STD FENDER HOMOLOGATION



MMLI

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# 1. Display Type and View

Describes how to view the display of this controller.

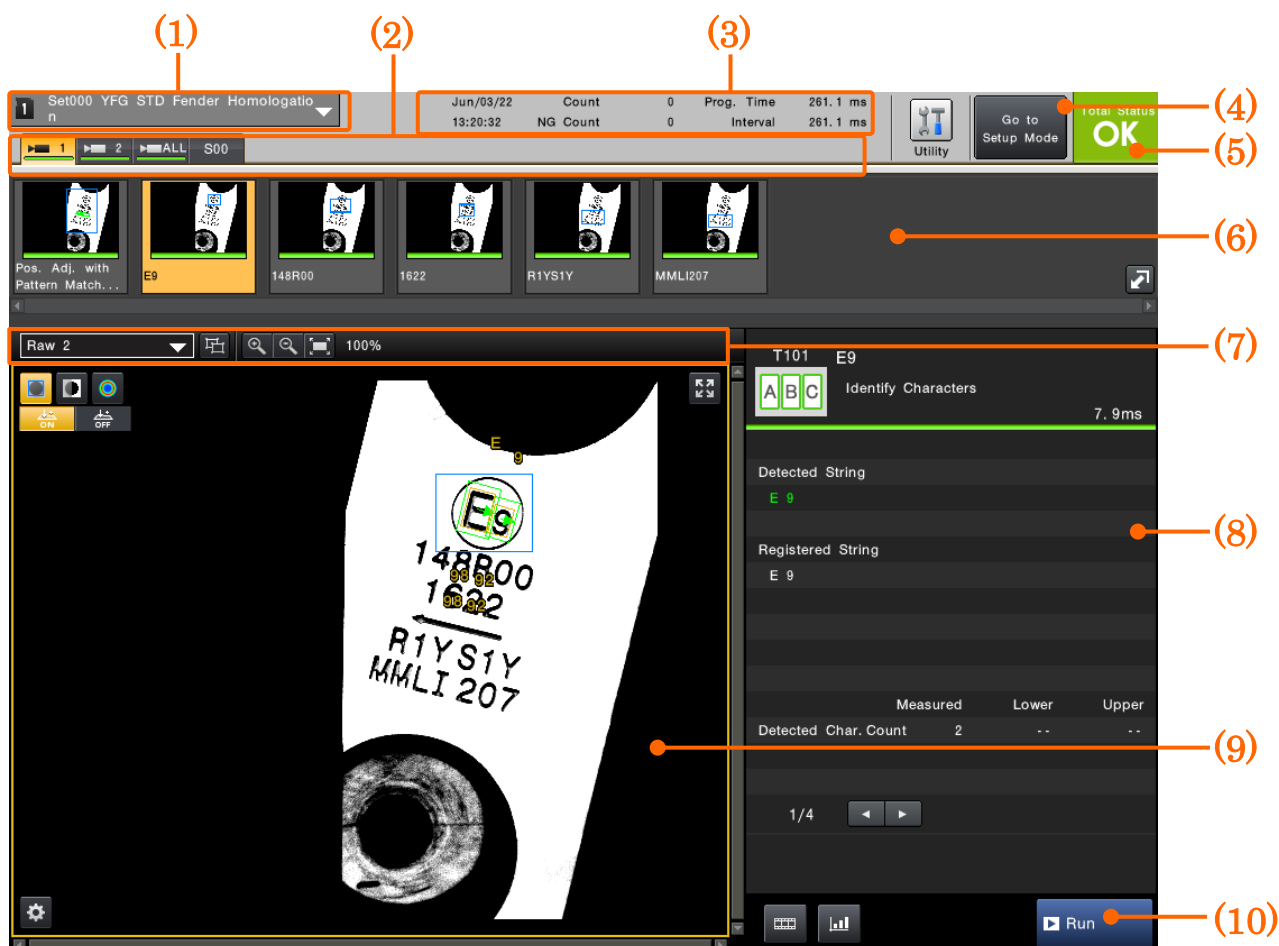
The display varies according to the difference between run and setup modes as well as the operation condition.

## 1.1. Operation Screen

The display shown in run mode.

### 1.1.1. Display View and Operation

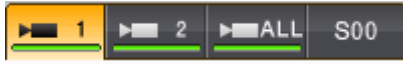
Items labelled with **Operation Target** come with an operation guide **Operation**. Refer to this guide.



### (1)Setting No./Name Display

Displays the setting No. and name currently in operation.

### (2)Display Tab Operation Target



These are the tabs used to switch between displays. There are three tabs: CAM tab, ALL tab and Custom tab.

Operation

Clicking the displayed tab switches the operation screen display. Refer to [1.1.2 Operation Screen List](#) for the list of screens that can be switched to in this setting.

### (3)Information Display Area

Displays Date, Total Count, Total Status NG Count, Program Time and Interval.

### (4)Mode Change Button (Run Mode/Setup Mode) Operation Target



Operation

Pressing the button can switch the mode between the run and setup modes.

### (5)Total Status Display



### (6)Toolbar (Tool List) Operation Target

Displays the list of the set tools in thumbnail format.

The toolbar thumbnails the tools used in the relevant camera for the CAM tab. For the ALL tab, all of the tools used are displayed.

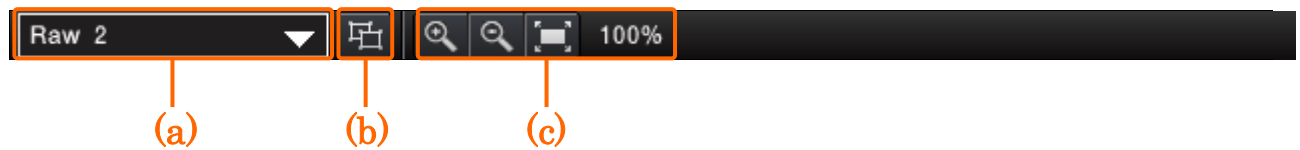
Operation

Click a thumbnail to change the region and measured value displayed in the camera screen to those of the clicked tool.

### (7)View Bar Operation Target

Zooms in/out the image displayed or changes the display type.

Operation



(a) Change Display Type

(b) Multiple Regions Display Icon



... Multiple Regions Display OFF



... Multiple Regions Display ON

(c) Display Magnification Icon



... Zoom in



... Zoom out



... Fit



... Current

### (8)Measured Value display

Displays the measured value of the selected tool.

### (9)Image Display

Displays the image input from the camera.

### (10)Run (trigger input) button Operation target

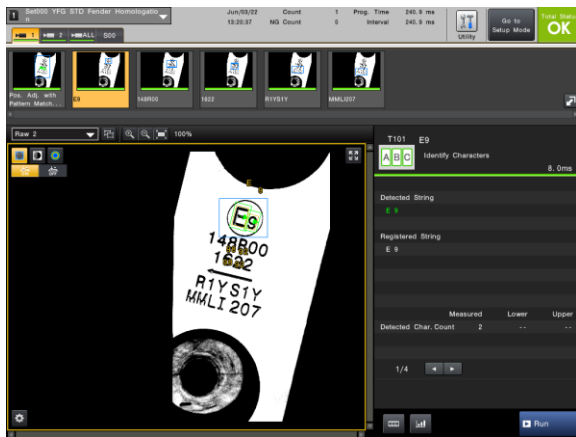
Operation

Press the button to run the measurement.

### 1.1.2. Operation Screen List

The following are the types screen which can be displayed during operation:

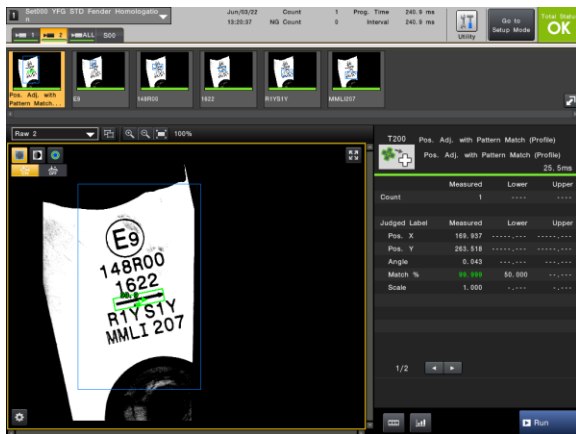
The inspection screen can be switched by clicking on the appropriate CAM tab.



CAM 1 tab

Inspection screen of CAM 1.

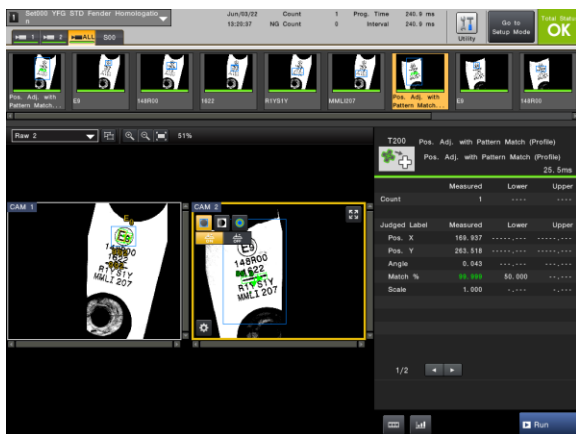
The tools linked with CAM 1 are displayed in the toolbar.



CAM 2 tab

Inspection screen of CAM 2.

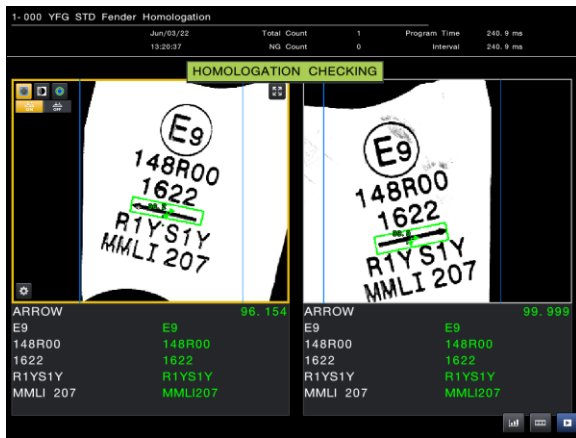
The tools linked with CAM 2 are displayed in the toolbar.



CAM ALL Tab

Inspection screen for all cameras.

The toolbar shows all the tools.



## S00 Tab

Operation screen S00 [Current Image for 2 Inspections].

The screen layout is as follows.

Display 0 CAM 1 Current Image

Display 1 CAM 2 Current Image

## 1.2. Setting Screen

The screen shown in setup mode.

### 1.2.1. Display View and Operation

The item with the **Operation Target** describes the **operation** of the operation guide. Refer to this guide.

Note that the description is omitted for the operation identical to the run mode.



**(1)Camera Setting Button** Operation Target

Operation

Press the button to show the camera setting screen.

**(2)Add Tool Button** Operation Target

Operation

Press the button to show the add tool screen.

**(3)Save Button** Operation Target

Operation

Press the button to save the current program setting.

**(4)Edit Button** Operation Target

Operation

Press the button to show the menu for editing the program setting.

**(5)Global Setting Button** Operation Target

Operation

Press the button to show the menu for editing the global setting.

**(6)Execute Condition Button** Operation Target

Operation

Press the button to show the execution condition setting screen.

**(7)Output Setting Button** Operation Target

Operation

Press the button to show the output setting screen.

**(8)Toolbar (Tool List)** Operation Target

Displays the list of the set tools in thumbnail format.

The toolbar thumbnails the tools used in the relevant camera for the CAM tab. For the ALL tab, all the tools used are displayed.

Operation

Select a thumbnail and press the “Edit” button (8'), and the tool specified by the thumbnail can be edited.

Double-clicking the thumbnail performs the same action.

**(9)Position Adjustment Button** Operation Target

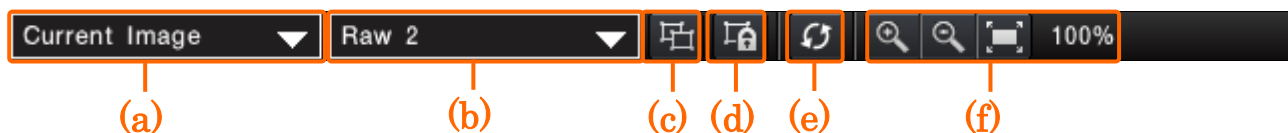
Operation

Press the button to show the position adjustment setting screen.

**(10)View Bar** Operation Target

Zooms in/out the image displayed or changes the display type.

Operation



(a) Change Display Image

(b) Change Display Type

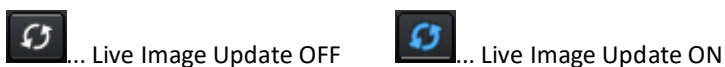
(c) Multiple Regions Display Icon



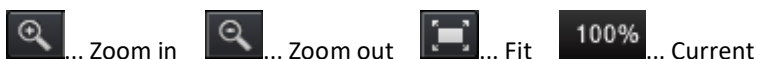
(d) Region Lock Icon



(e) Live Image Icon



(f) Display Magnification Icon



**(11)Image Strip Button** Operation Target

Operation

Pressing the button can display or hide the image strip.

**(12)Image Registration Button** Operation Target

Operation

Press the button to show the image registration display.

**(13)Repeat Execution Button** Operation Target

Operation

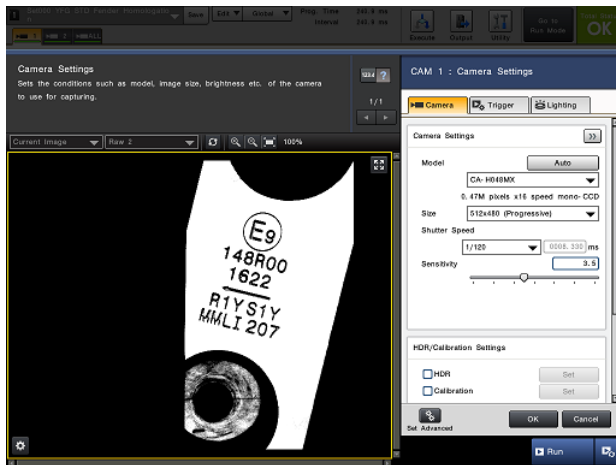
Press the button to execute continuous measurement.

## 2. Camera Adjustment

### 2.1. (CAM1) Camera Settings

Sets the conditions such as model, image size, brightness etc. of the camera to use for capturing.

#### 2.1.1. Defaults List of Camera Parameters



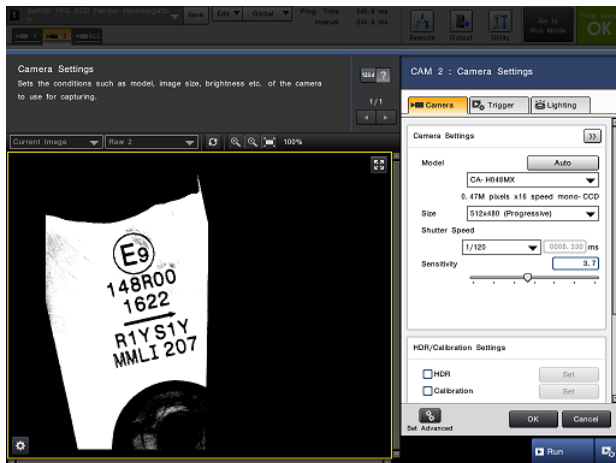
[Camera Settings]

| Parameter Name | Defaults                  | On Change | Memo |
|----------------|---------------------------|-----------|------|
| Model          | CA-H048MX                 |           |      |
| Size           | 512x480<br>(Progressive)  |           |      |
| Shutter Speed  | 1/120 (8.33ms)            |           |      |
| Sensitivity    | 3.5                       |           |      |
| Shift          | -170                      |           |      |
| Span           | 7.9                       |           |      |
| Mode           | Standard Lighting<br>Mode |           |      |
| HDR            | Disabled                  |           |      |
| Calibration    | Disabled                  |           |      |

### 2.2. (CAM2) Camera Settings

Sets the conditions such as model, image size, brightness etc. of the camera to use for capturing.

### 2.2.1. Defaults List of Camera Parameters



#### [Camera Settings]

| Parameter Name | Defaults                  | On Change | Memo |
|----------------|---------------------------|-----------|------|
| Model          | CA-H048MX                 |           |      |
| Size           | 512x480<br>(Progressive)  |           |      |
| Shutter Speed  | 1/120 (8.33ms)            |           |      |
| Sensitivity    | 3.7                       |           |      |
| Shift          | -130                      |           |      |
| Span           | 7.9                       |           |      |
| Mode           | Standard Lighting<br>Mode |           |      |
| HDR            | Disabled                  |           |      |
| Calibration    | Disabled                  |           |      |

---

## 3. Tool Adjustment

---

By changing to [Setup Mode] using the mode change button, setting adjustment of each tool can be done.

### 3.1. Tool List

The following tools are used for this setting:

| CAM 1 |                                        |      |           |
|-------|----------------------------------------|------|-----------|
| ID    | Tool Name                              | Memo | Reference |
| T100  | Pos. Adj. with Pattern Match (Profile) |      | p.19      |
| T101  | E9                                     |      | p.25      |
| T102  | 148R00                                 |      | p.32      |
| T103  | 1622                                   |      | p.39      |
| T104  | R1YS1Y                                 |      | p.46      |
| T105  | MMLI207                                |      | p.53      |
| CAM 2 |                                        |      |           |
| ID    | Tool Name                              | Memo | Reference |
| T200  | Pos. Adj. with Pattern Match (Profile) |      | p.60      |
| T201  | E9                                     |      | p.66      |
| T202  | 148R00                                 |      | p.73      |
| T203  | 1622                                   |      | p.80      |
| T204  | R1YS1Y                                 |      | p.87      |
| T205  | MMLI207                                |      | p.94      |

## 3.2. (CAM1) [T100]Pos. Adj. with Pattern Match (Profile)



### 3.2.1. Summary

#### Summary of this tool

Detects the most similar portion to the profile information registered in advance and outputs the position and angle of the detected target object.

This tool is applied as a position correction reference for another tool (target) where position adjustment is performed according to the pattern position and angle gap information detected with this tool.

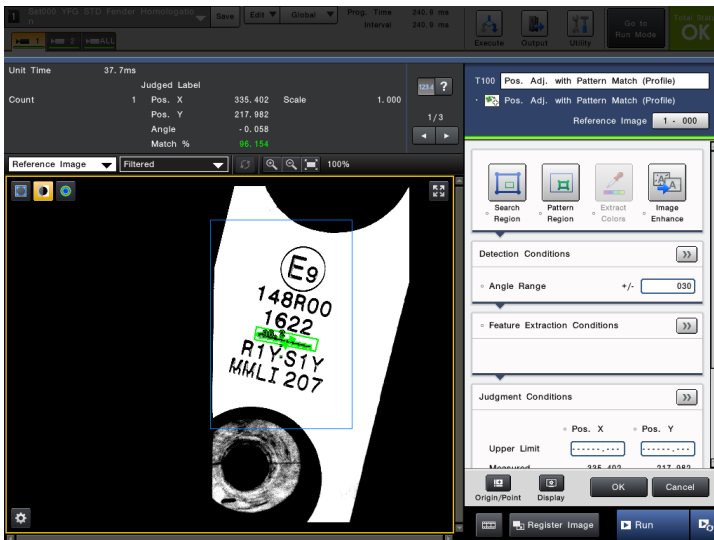
#### Memo

### 3.2.2. Defaults List of Parameters

| Parameter Name    | Defaults | On Change | Memo |
|-------------------|----------|-----------|------|
| Angle Range (+/-) | 30       |           |      |

### 3.2.3. Tool Edit Screen View

This is the edit screen for the [T100]Pos. Adj. with Pattern Match (Profile) tool.



#### [Search Region]

Sets the search region. For details, refer to [3.2.4 Setting the Search Region](#).



#### [Pattern Region]

Sets the pattern region. For details, refer to [3.2.5 Setting the Pattern Region](#).



#### [Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

#### [Detection Conditions]

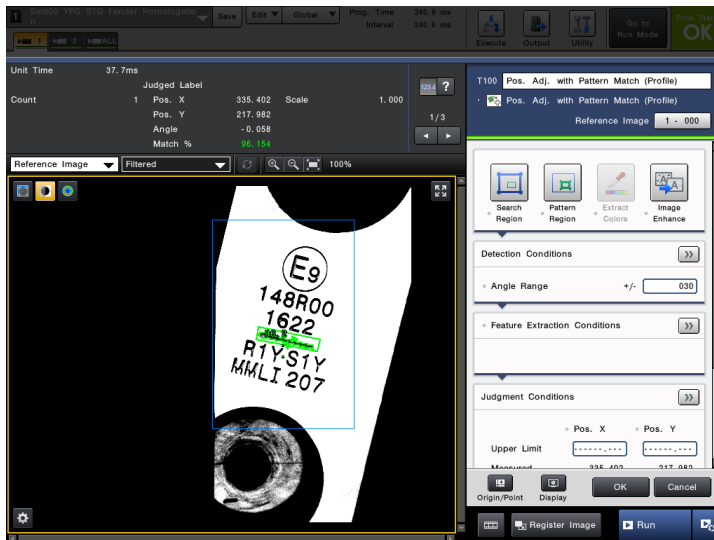
Sets the detection conditions according to the measurement items. For details, refer to [3.2.6 Adjustment for Detection Conditions](#).

#### [Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.2.7 Adjustment for Judgment Conditions](#).

### 3.2.4. Setting the Search Region

#### 3.2.4.1. Search Region



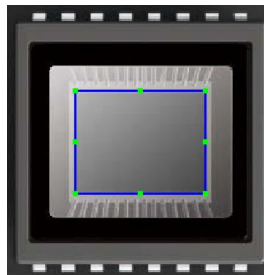
Specifies the region to be measured.

Rectangle is used for specifying the search region in this particular case.

#### [Rectangle]

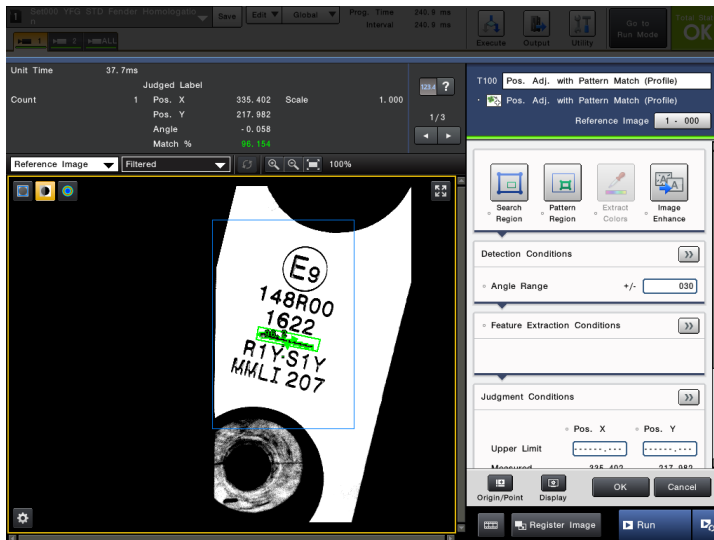
(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



## 3.2.5. Setting the Pattern Region

### 3.2.5.1. Pattern Region



Specifies the characteristic pattern (shape) to be detected by setting the pattern region.

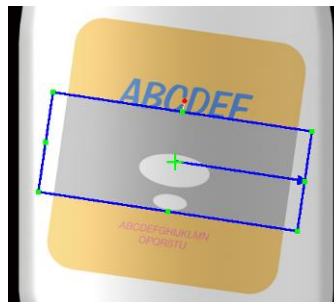
Rotated Rectangle is used for specifying the pattern region in this particular case.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



## 3.2.6. Adjustment for Detection Conditions

### 3.2.6.1. Angle Range

| Detection Conditions |                                      |
|----------------------|--------------------------------------|
| ◦ Angle Range        | +/- <input type="text" value="030"/> |

Specifies the angle range by +/- for the case where the search target is titled.

#### [Angle Range]

##### (General Overview)

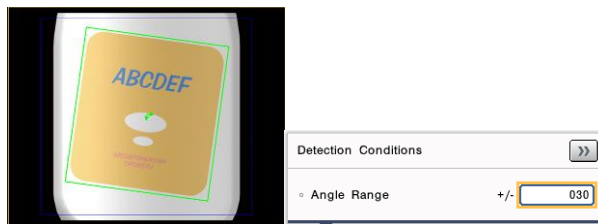
This parameter specifies an angle range corresponding to the rotation of a pattern.



When the angle range is 0, the rotated object cannot be detected.

##### (Usage)

If the target possibly rotates, specify a prospective angle range such that rotated targets can be detected.



Specify the angle range, and the rotated object can be detected.

### 3.2.7. Adjustment for Judgment Conditions

Judgment Conditions

Pos. X

Pos. Y

Upper Limit

Measured

Lower Limit

335.402

217.982

Angle

Match %

Upper Limit

Measured

Lower Limit

- 0.058

96.154

50.000

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Match %

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

#### 3.2.7.1. Defaults List of Judgment Conditions

| Judgment Conditions Name | Defaults    |             | On Change   |             | Memo |
|--------------------------|-------------|-------------|-------------|-------------|------|
|                          | Upper Limit | Lower Limit | Upper Limit | Lower Limit |      |
| Pos. X                   | -----       | -----       |             |             |      |
| Pos. Y                   | -----       | -----       |             |             |      |
| Angle                    | -----       | -----       |             |             |      |
| Match %                  | -----       | 50          |             |             |      |

### 3.3. (CAM1) [T101]E9



#### 3.3.1. Summary

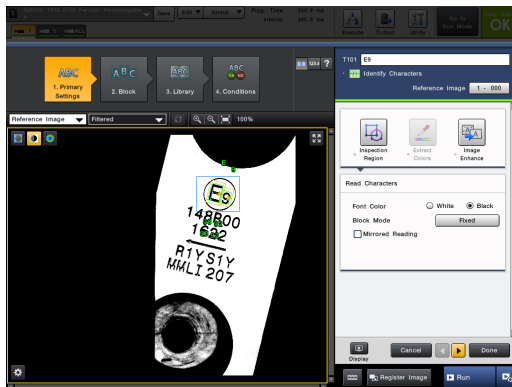
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

#### 3.3.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.3.3. Tool Edit Screen View

This is the edit screen for the [T101]E9 tool.



#### [Step Buttons]

This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|         |                  |
|---------|------------------|
| Step. 1 | Primary Settings |
|---------|------------------|

- Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.

Note: For details, refer to [3.3.4 Step. 1 Primary Settings](#).

|         |       |
|---------|-------|
| Step. 2 | Block |
|---------|-------|

- Sets to block the character.

Note: For details, refer to [3.3.6 Step. 2 Block](#).

|         |                  |
|---------|------------------|
| Step. 3 | Library Settings |
|---------|------------------|

- Registration to the library is made for the character pattern used in the recognize settings.

Note: For details, refer to [3.3.7 Step. 3 Library](#).

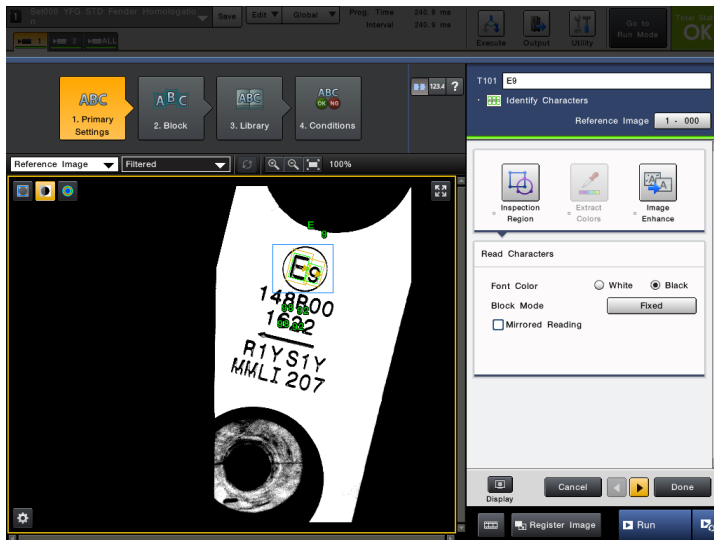
|         |           |
|---------|-----------|
| Step. 4 | Condition |
|---------|-----------|

- Sets the judgment conditions with OK/NG-judgment to the recognized data.

Note: For details, refer to [3.3.8 Step. 4 Condition](#).

### 3.3.4. Step. 1 Primary Settings

This is the edit screen for the [T101]E9 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.3.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

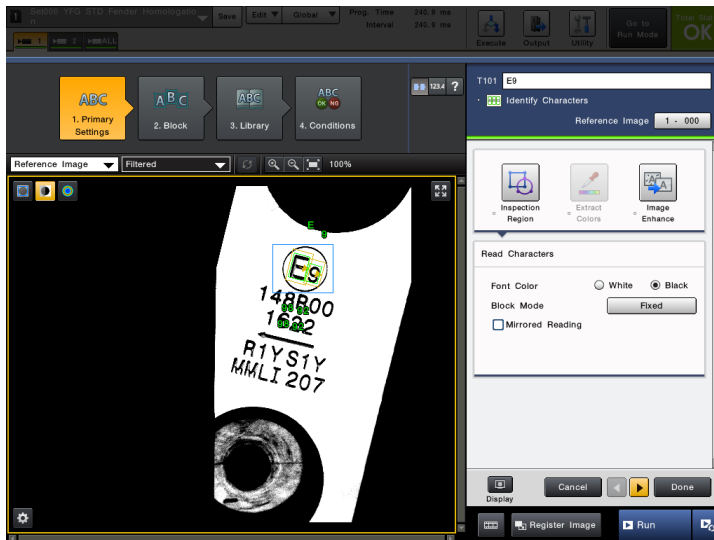
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.3.5. Setting the Inspection Region

#### 3.3.5.1. Inspection Region



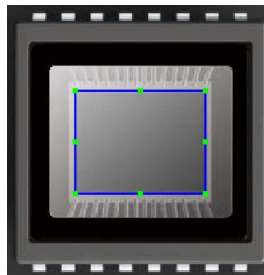
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

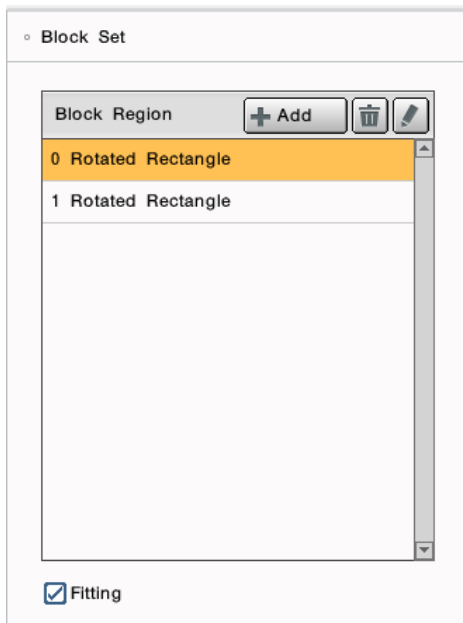
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.3.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

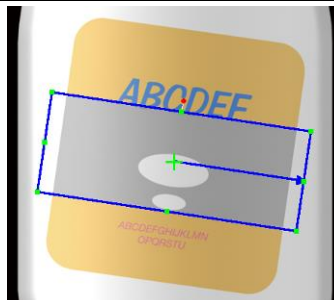
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.3.7. Step. 3 Library

Library Settings

☐ Built-in Library
☒ Library No.

Create Library
Edit

| Library Name                 | New Library      |
|------------------------------|------------------|
| Registered Characters        | 210 Character(s) |
| Upper Case                   | 7 Type 80 Char.  |
| Lower Case                   | None             |
| Number                       | 9 Type 130 Char. |
| Symbol                       | None             |
| (Built-in Library Characters | 166Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#) .

### 3.3.8. Step. 4 Condition

Judgment Conditions

Detected String  
E9

Registered String

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 98                              |
| Minimum     | 92                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.3.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | E9       |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.4. (CAM1) [T102]148R00



#### 3.4.1. Summary

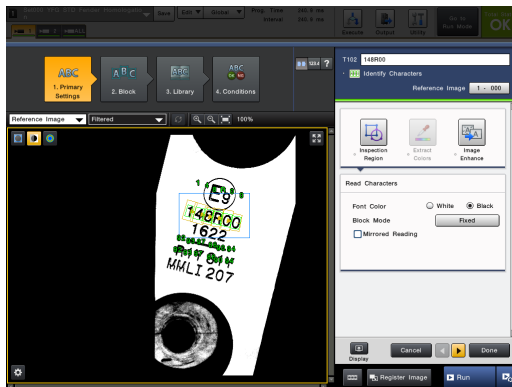
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

#### 3.4.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.4.3. Tool Edit Screen View

This is the edit screen for the [T102]148R00 tool.



#### [Step Buttons]

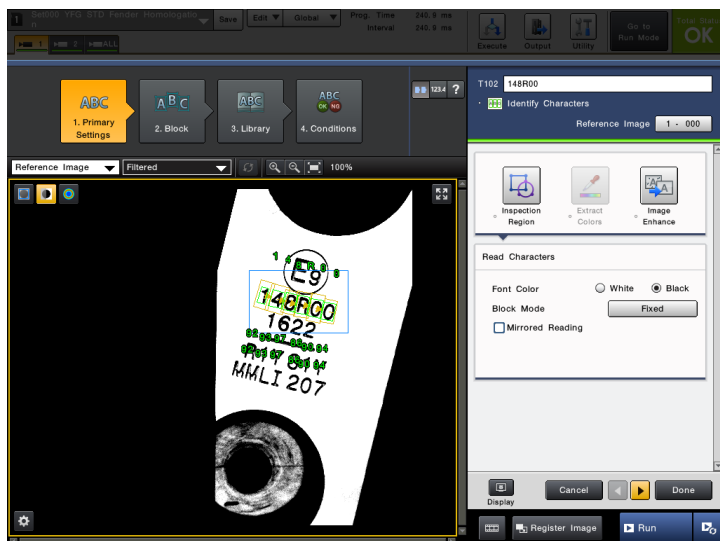
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                             |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                     | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.4.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                     | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.4.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                     | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.4.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                     | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.4.8 Step. 4 Condition</a> .                                         |                  |

### 3.4.4. Step. 1 Primary Settings

This is the edit screen for the [T102]148R00 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.4.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

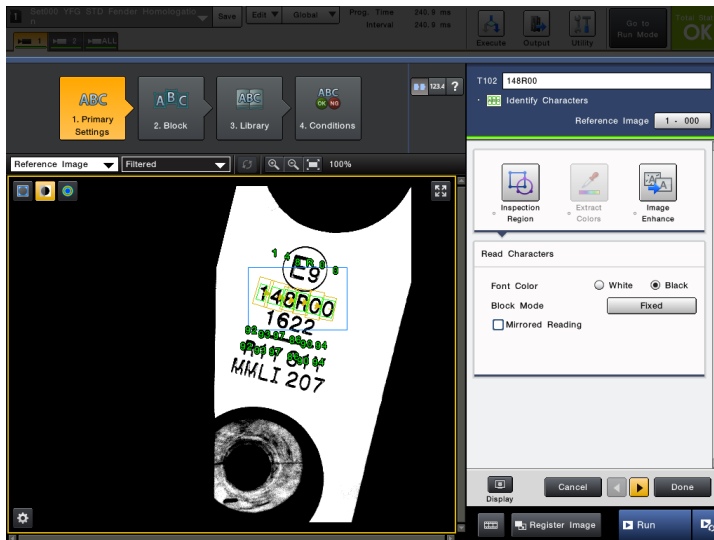
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.4.5. Setting the Inspection Region

#### 3.4.5.1. Inspection Region



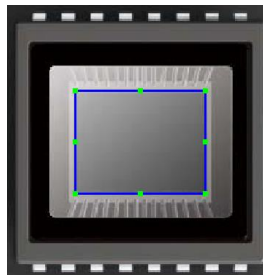
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

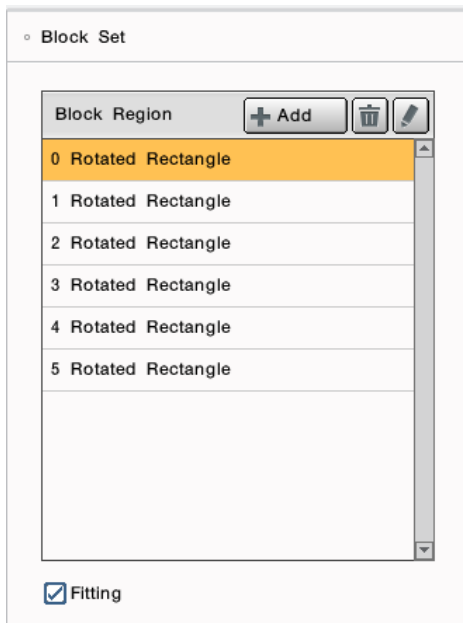
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.4.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

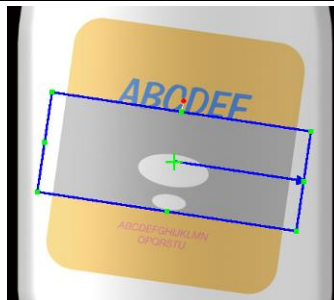
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.4.7. Step. 3 Library

Library Settings

☐ Built-in Library

☒ Library No.

Create Library

Edit

| Library Name                                  | New Library      |
|-----------------------------------------------|------------------|
| Registered Characters                         | 210 Character(s) |
| Upper Case                                    | 7 Type 80 Char.  |
| Lower Case                                    | None             |
| Number                                        | 9 Type 130 Char. |
| Symbol                                        | None             |
| (Built-in Library Characters 166Character(s)) |                  |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.4.8. Step. 4 Condition

Judgment Conditions

Detected String  
148R00

Registered String

Copy Detected String

Calendar Settings

Restriction Settings

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 97                              |
| Minimum     | 86                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Matching Settings

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.4.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | 148R00   |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.5. (CAM1) [T103]1622



### 3.5.1. Summary

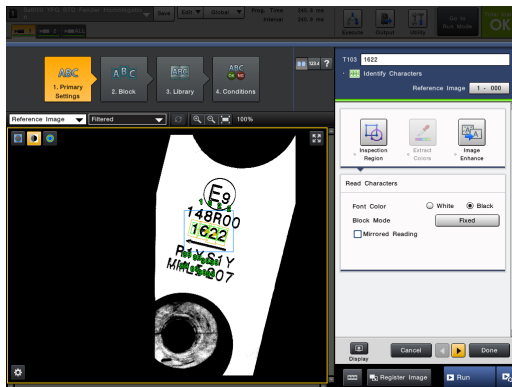
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.5.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.5.3. Tool Edit Screen View

This is the edit screen for the [T103]1622 tool.



#### [Step Buttons]

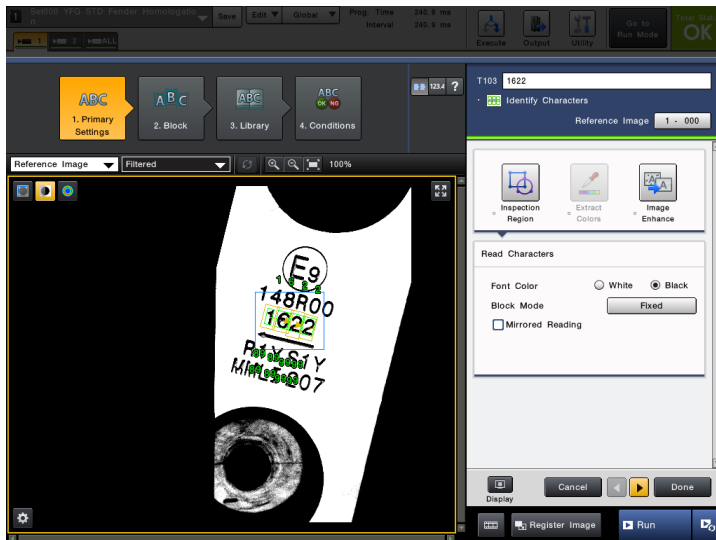
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                             |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                     | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.5.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                     | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.5.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                     | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.5.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                     | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.5.8 Step. 4 Condition</a> .                                         |                  |

### 3.5.4. Step. 1 Primary Settings

This is the edit screen for the [T103]1622 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.5.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

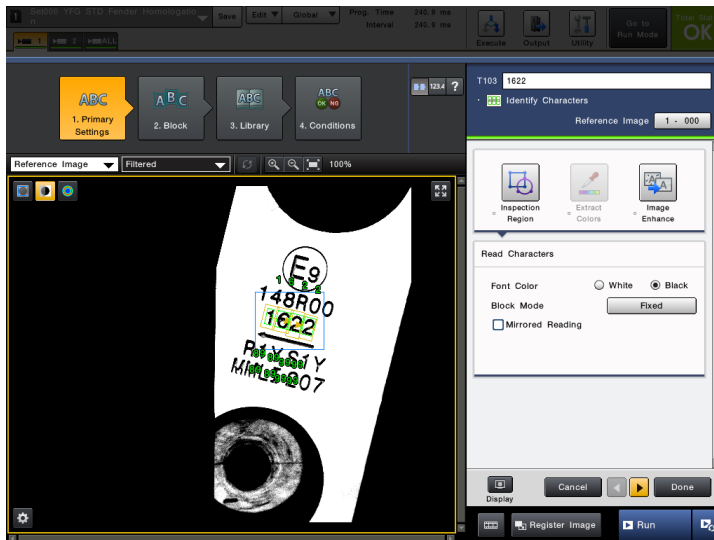
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.5.5. Setting the Inspection Region

#### 3.5.5.1. Inspection Region



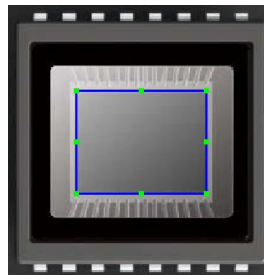
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

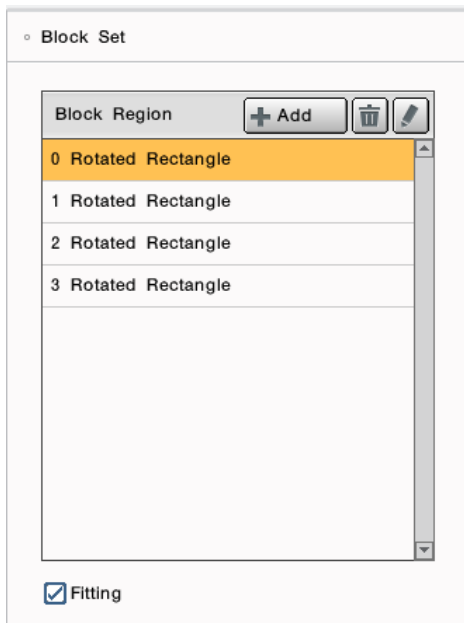
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.5.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

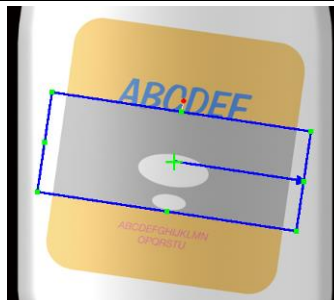
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.5.7. Step. 3 Library

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.5.8. Step. 4 Condition

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.5.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | 1622     |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.6. (CAM1) [T104]R1YS1Y



### 3.6.1. Summary

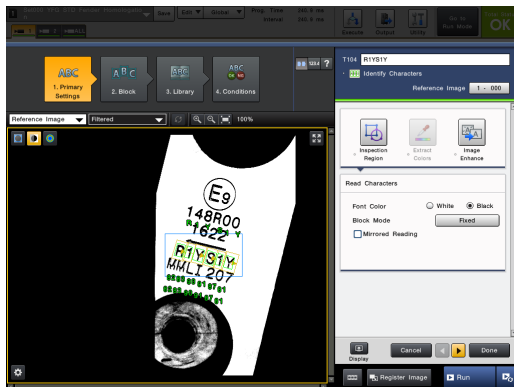
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.6.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.6.3. Tool Edit Screen View

This is the edit screen for the [T104]R1YS1Y tool.



#### [Step Buttons]

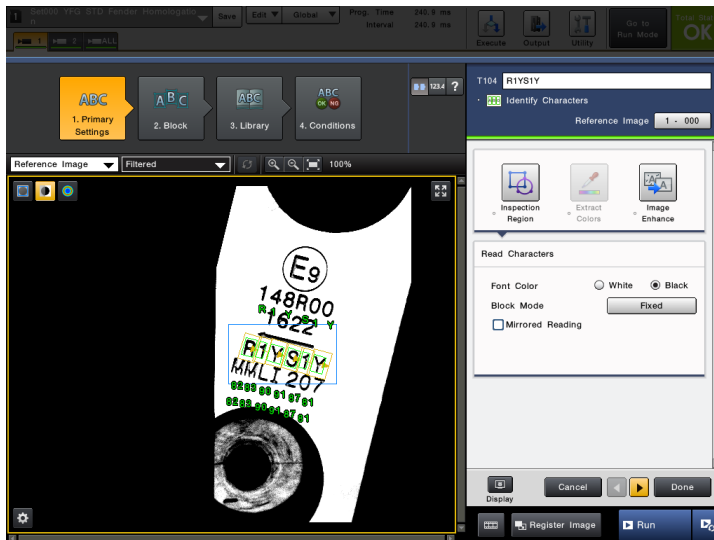
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                             |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                     | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.6.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                     | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.6.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                     | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.6.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                     | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.6.8 Step. 4 Condition</a> .                                         |                  |

### 3.6.4. Step. 1 Primary Settings

This is the edit screen for the [T104]R1YS1Y tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.6.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

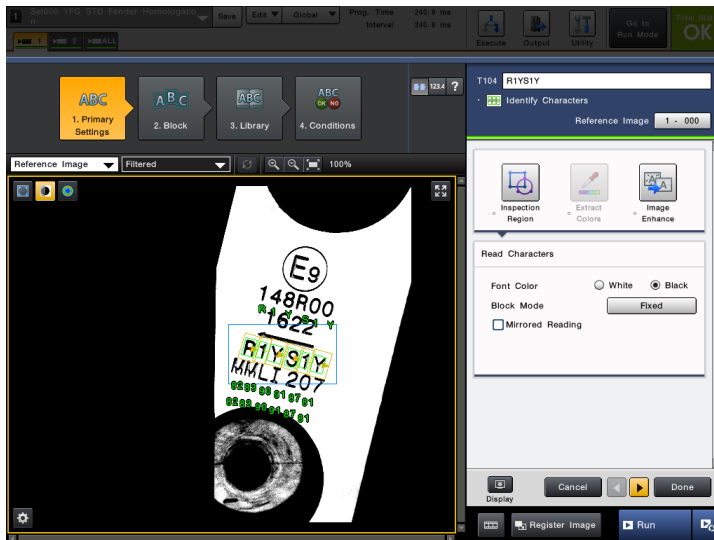
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.6.5. Setting the Inspection Region

#### 3.6.5.1. Inspection Region



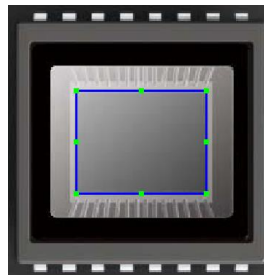
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

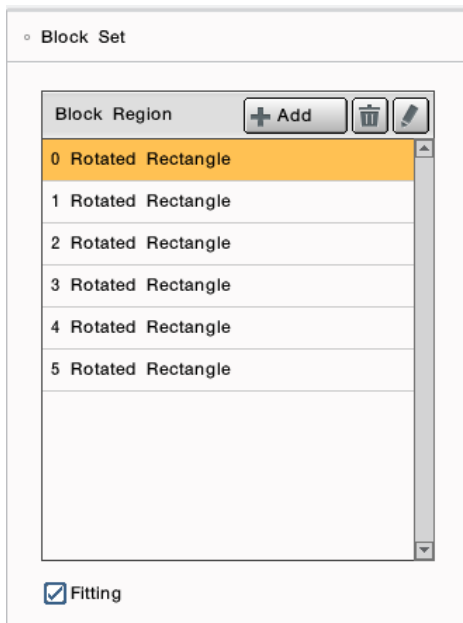
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.6.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the  button.

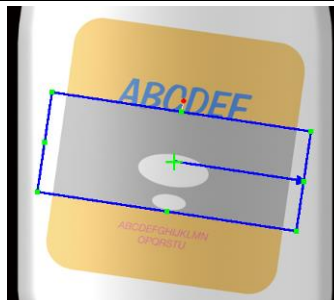
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.6.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library      |
|------------------------------|------------------|
| Registered Characters        | 210 Character(s) |
| Upper Case                   | 7 Type 80 Char.  |
| Lower Case                   | None             |
| Number                       | 9 Type 130 Char. |
| Symbol                       | None             |
| (Built-in Library Characters | 166Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.6.8. Step. 4 Condition

Judgment Conditions

Detected String  
R1YS1Y

Registered String

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 91                              |
| Minimum     | 81                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.6.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | R1YS1Y   |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.7. (CAM1) [T105]MMLI207



### 3.7.1. Summary

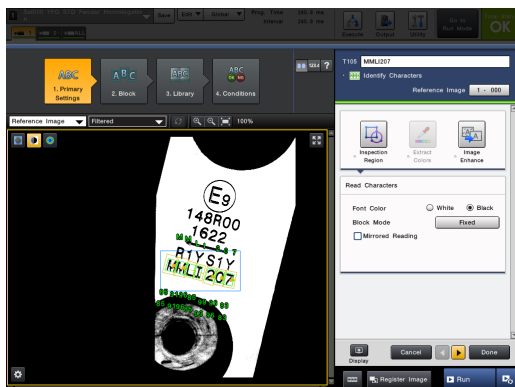
| Summary of this tool                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.7.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.7.3. Tool Edit Screen View

This is the edit screen for the [T105]MMLI207 tool.



#### [Step Buttons]

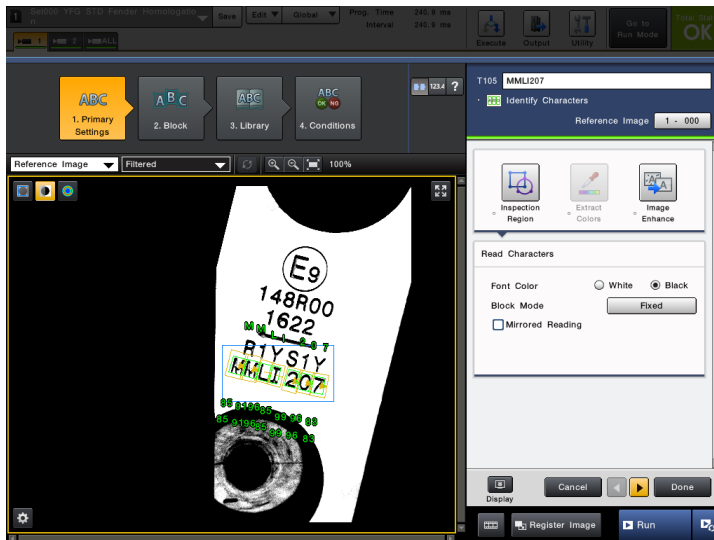
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                             |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                     | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.7.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                     | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.7.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                     | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.7.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                     | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.7.8 Step. 4 Condition</a> .                                         |                  |

### 3.7.4. Step. 1 Primary Settings

This is the edit screen for the [T105]MMLI207 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.7.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

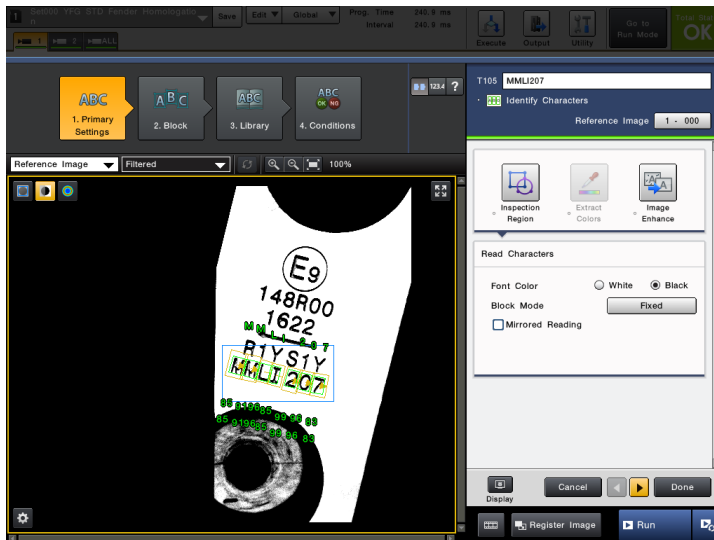
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.7.5. Setting the Inspection Region

#### 3.7.5.1. Inspection Region



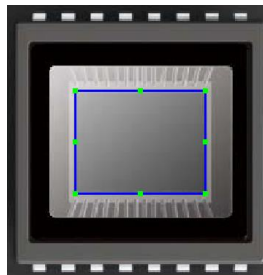
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

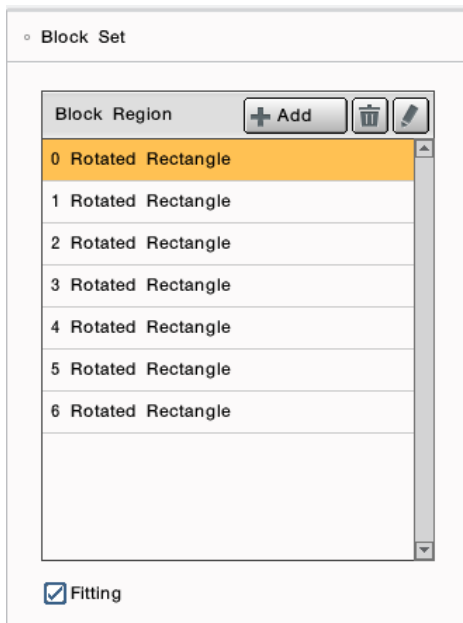
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.7.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

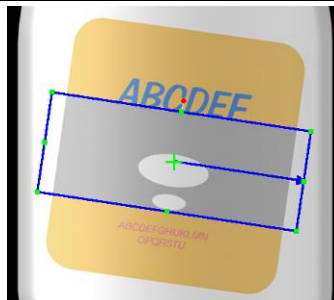
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.7.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library       |
|------------------------------|-------------------|
| Registered Characters        | 210 Character(s)  |
| Upper Case                   | 7 Type 80 Char.   |
| Lower Case                   | None              |
| Number                       | 9 Type 130 Char.  |
| Symbol                       | None              |
| (Built-in Library Characters | 166 Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.7.8. Step. 4 Condition

Judgment Conditions

Detected String  
MMLI207

Registered String

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 99                              |
| Minimum     | 83                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.7.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | MMLI207  |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

3.8. (CAM2) [T200]Pos. Adj. with Pattern Match (Profile)



3.8.1. Summary

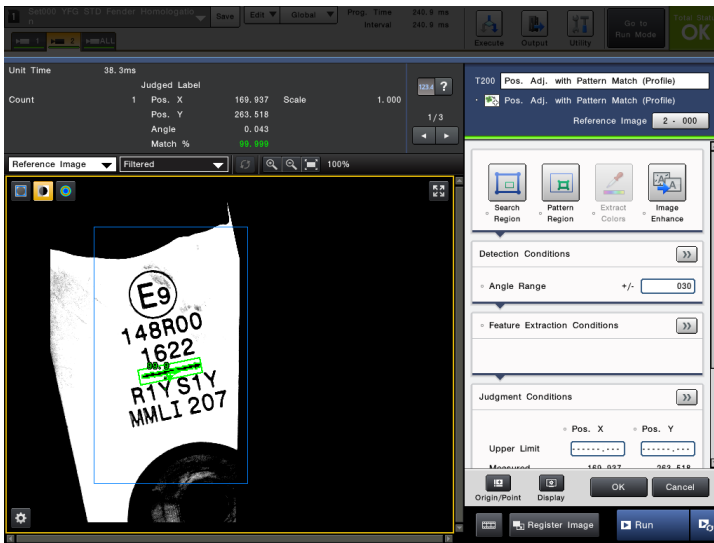
| Summary of this tool                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Detects the most similar portion to the profile information registered in advance and outputs the position and angle of the detected target object.</p> <p>This tool is applied as a position correction reference for another tool (target) where position adjustment is performed according to the pattern position and angle gap information detected with this tool.</p> |
| Memo                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                 |

3.8.2. Defaults List of Parameters

| Parameter Name    | Defaults | On Change | Memo |
|-------------------|----------|-----------|------|
| Angle Range (+/-) | 30       |           |      |

### 3.8.3. Tool Edit Screen View

This is the edit screen for the [T200]Pos. Adj. with Pattern Match (Profile) tool.



#### [Search Region]

Sets the search region. For details, refer to [3.8.4 Setting the Search Region](#).



#### [Pattern Region]

Sets the pattern region. For details, refer to [3.8.5 Setting the Pattern Region](#).



#### [Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

#### [Detection Conditions]

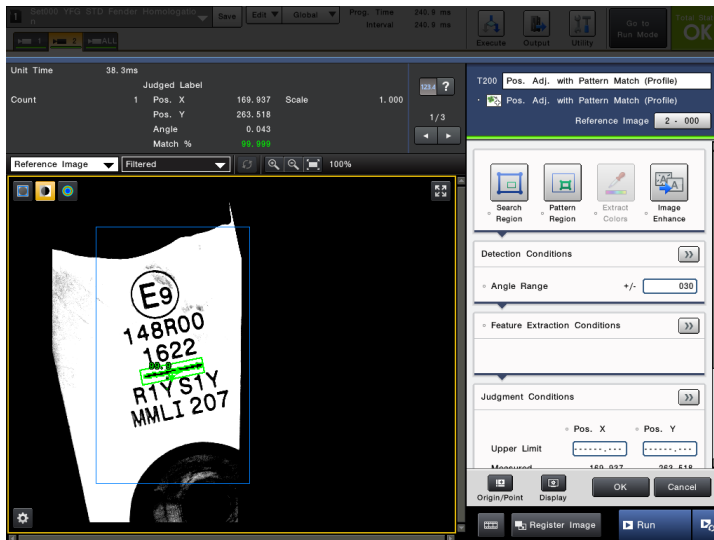
Sets the detection conditions according to the measurement items. For details, refer to [3.8.6 Adjustment for Detection Conditions](#).

#### [Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.8.7 Adjustment for Judgment Conditions](#).

### 3.8.4. Setting the Search Region

#### 3.8.4.1. Search Region



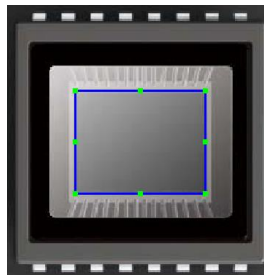
Specifies the region to be measured.

Rectangle is used for specifying the search region in this particular case.

#### [Rectangle]

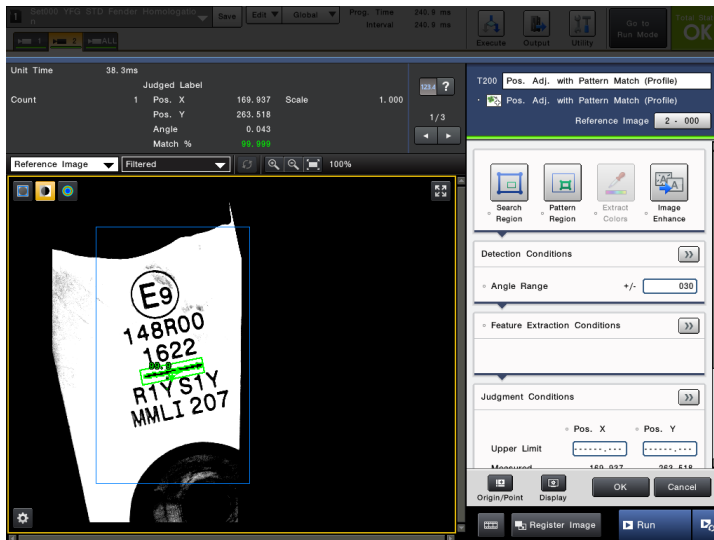
(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



## 3.8.5. Setting the Pattern Region

### 3.8.5.1. Pattern Region



Specifies the characteristic pattern (shape) to be detected by setting the pattern region.

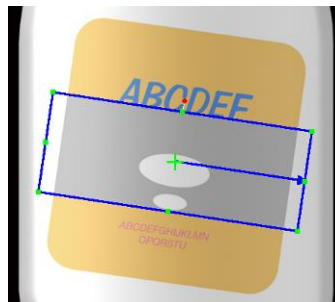
Rotated Rectangle is used for specifying the pattern region in this particular case.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



## 3.8.6. Adjustment for Detection Conditions

### 3.8.6.1. Angle Range

| Detection Conditions |     | >>                               |
|----------------------|-----|----------------------------------|
| ◦ Angle Range        | +/- | <input type="text" value="030"/> |

Specifies the angle range by +/- for the case where the search target is titled.

#### [Angle Range]

##### (General Overview)

This parameter specifies an angle range corresponding to the rotation of a pattern.

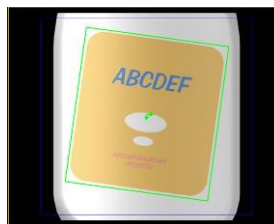


| Detection Conditions |     | >>                               |
|----------------------|-----|----------------------------------|
| ◦ Angle Range        | +/- | <input type="text" value="000"/> |

When the angle range is 0, the rotated object cannot be detected.

##### (Usage)

If the target possibly rotates, specify a prospective angle range such that rotated targets can be detected.



| Detection Conditions |     | >>                               |
|----------------------|-----|----------------------------------|
| ◦ Angle Range        | +/- | <input type="text" value="030"/> |

Specify the angle range, and the rotated object can be detected.

### 3.8.7. Adjustment for Judgment Conditions

Judgment Conditions

Pos. X

Pos. Y

Upper Limit

Measured

Lower Limit

Angle

Match %

Upper Limit

Measured

Lower Limit

-----

169.937

-----

-----

263.518

-----

-----

0.043

99.999

-----

50.000

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Match %

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

#### 3.8.7.1. Defaults List of Judgment Conditions

| Judgment Conditions Name | Defaults    |             | On Change   |             | Memo |
|--------------------------|-------------|-------------|-------------|-------------|------|
|                          | Upper Limit | Lower Limit | Upper Limit | Lower Limit |      |
| Pos. X                   | -----       | -----       |             |             |      |
| Pos. Y                   | -----       | -----       |             |             |      |
| Angle                    | -----       | -----       |             |             |      |
| Match %                  | -----       | 50          |             |             |      |

## 3.9. (CAM2) [T201]E9



### 3.9.1. Summary

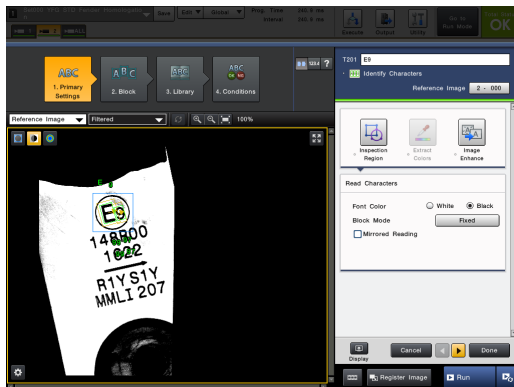
| Summary of this tool                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.9.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.9.3. Tool Edit Screen View

This is the edit screen for the [T201]E9 tool.



#### [Step Buttons]

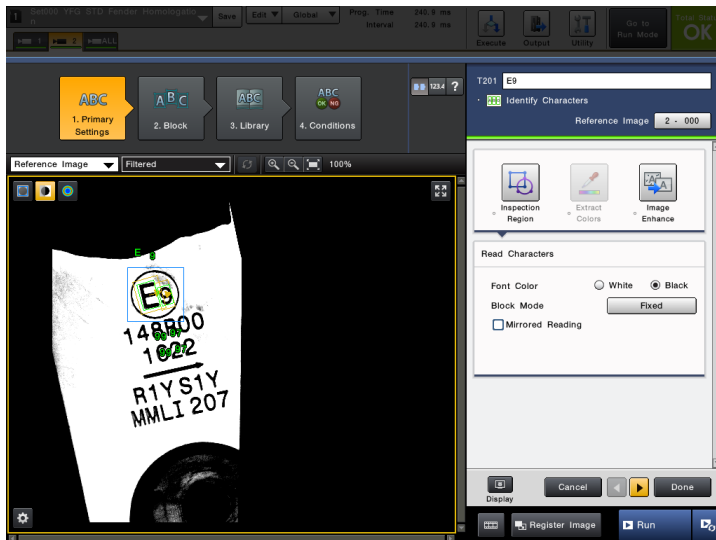
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                             |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                     | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.9.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                     | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.9.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                     | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.9.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                     | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.9.8 Step. 4 Condition</a> .                                         |                  |

### 3.9.4. Step. 1 Primary Settings

This is the edit screen for the [T201]E9 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.9.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

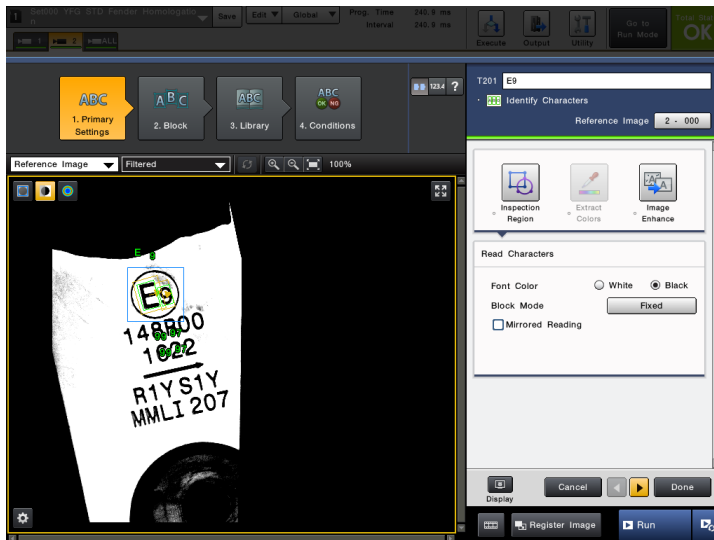
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.9.5. Setting the Inspection Region

#### 3.9.5.1. Inspection Region



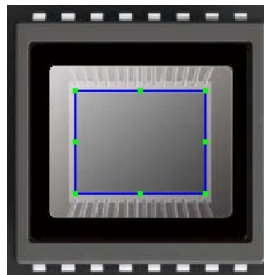
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

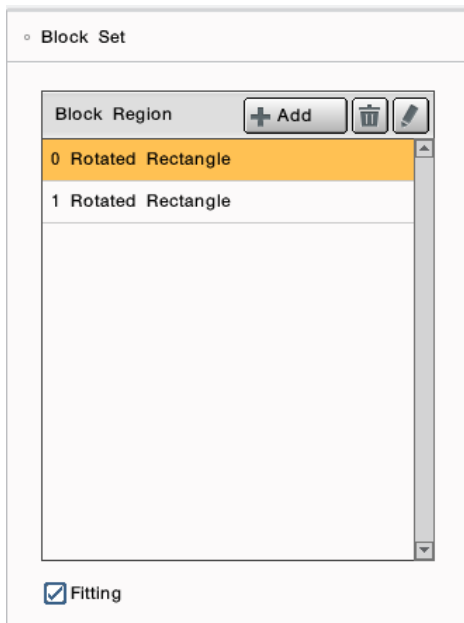
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.9.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

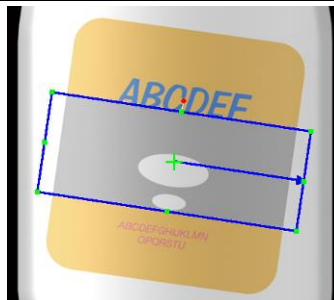
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.9.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library      |
|------------------------------|------------------|
| Registered Characters        | 210 Character(s) |
| Upper Case                   | 7 Type 80 Char.  |
| Lower Case                   | None             |
| Number                       | 9 Type 130 Char. |
| Symbol                       | None             |
| (Built-in Library Characters | 166Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.9.8. Step. 4 Condition

Judgment Conditions

Detected String  
E9

Registered String

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 99                              |
| Minimum     | 97                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.9.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | E9       |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.10. (CAM2) [T202]148R00



### 3.10.1. Summary

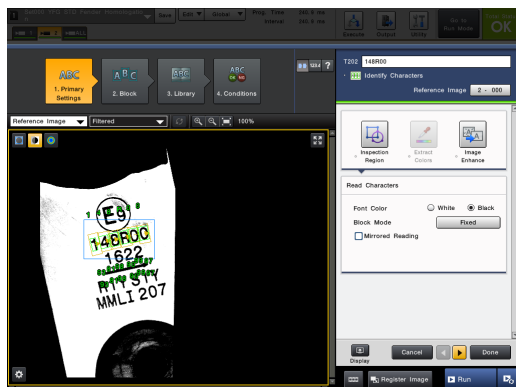
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.10.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.10.3. Tool Edit Screen View

This is the edit screen for the [T202]148R00 tool.



#### [Step Buttons]

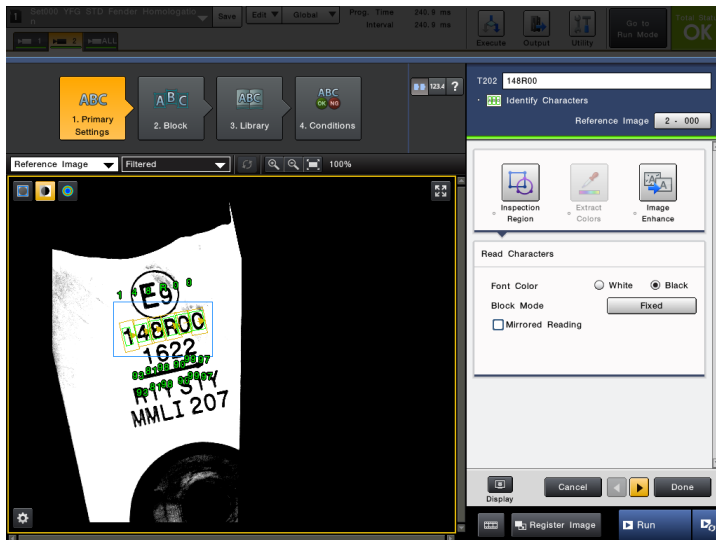
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                              |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                      | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.10.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                      | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.10.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                      | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.10.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                      | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.10.8 Step. 4 Condition</a> .                                         |                  |

### 3.10.4. Step. 1 Primary Settings

This is the edit screen for the [T202]148R00 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.10.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

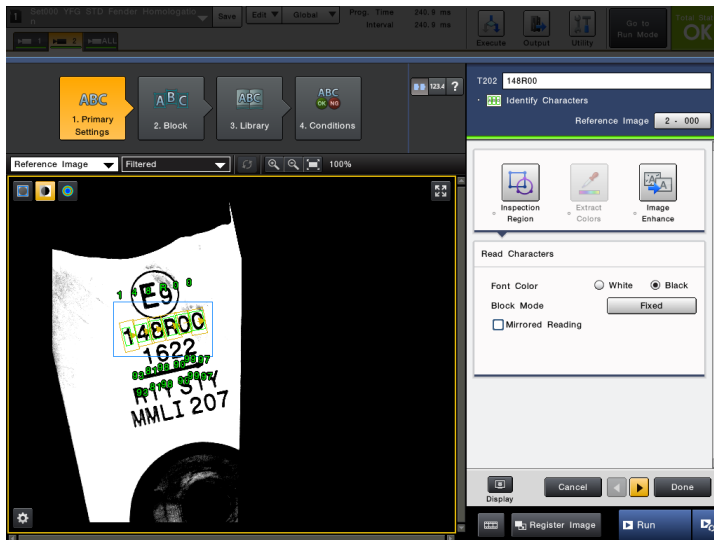
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.10.5. Setting the Inspection Region

#### 3.10.5.1. Inspection Region



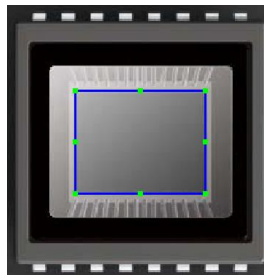
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

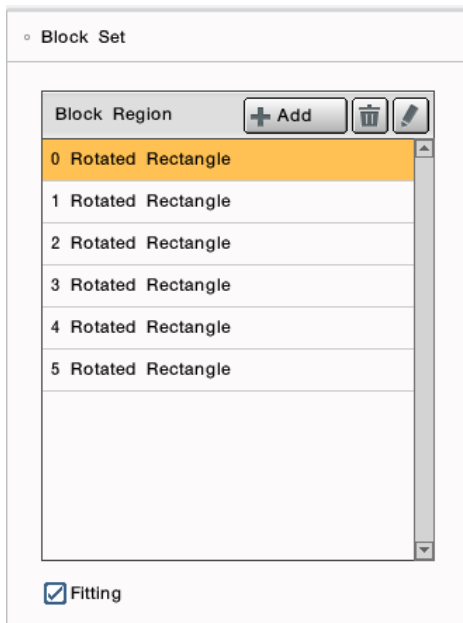
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.10.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

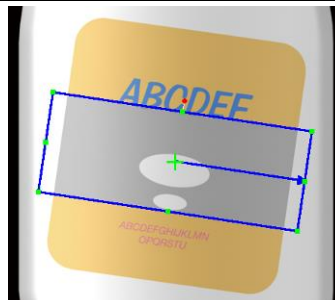
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.10.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library      |
|------------------------------|------------------|
| Registered Characters        | 210 Character(s) |
| Upper Case                   | 7 Type 80 Char.  |
| Lower Case                   | None             |
| Number                       | 9 Type 130 Char. |
| Symbol                       | None             |
| (Built-in Library Characters | 166Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.10.8. Step. 4 Condition

Judgment Conditions

Detected String  
148R00

Registered String

Correlation

Maximum 99

Minimum 81

Lower Limit

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.10.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | 148R00   |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.11. (CAM2) [T203]1622



### 3.11.1. Summary

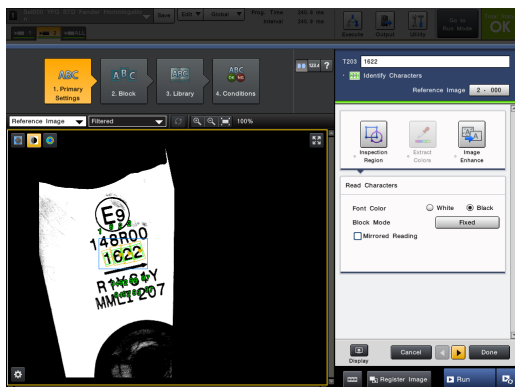
| Summary of this tool                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.11.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.11.3. Tool Edit Screen View

This is the edit screen for the [T203]1622 tool.



#### [Step Buttons]

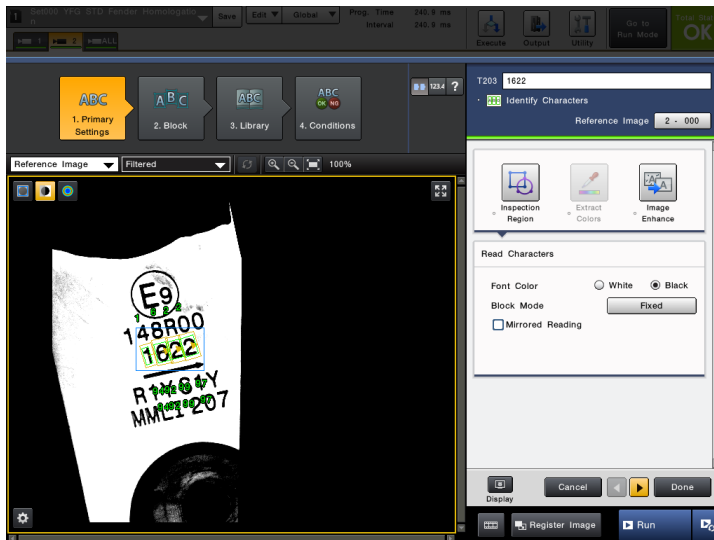
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                              |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                      | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.11.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                      | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.11.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                      | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.11.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                      | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.11.8 Step. 4 Condition</a> .                                         |                  |

### 3.11.4. Step. 1 Primary Settings

This is the edit screen for the [T203]1622 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.11.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

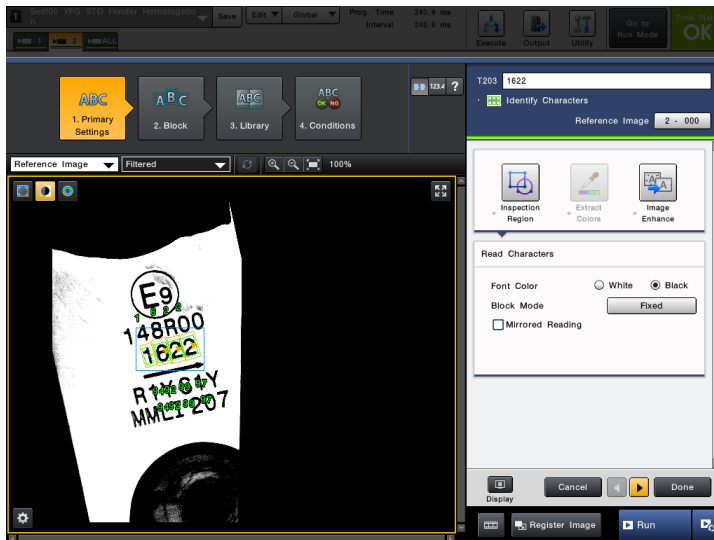
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.11.5. Setting the Inspection Region

#### 3.11.5.1. Inspection Region



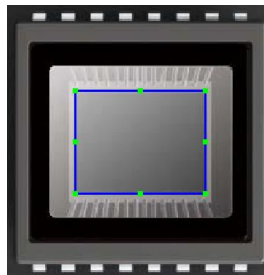
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

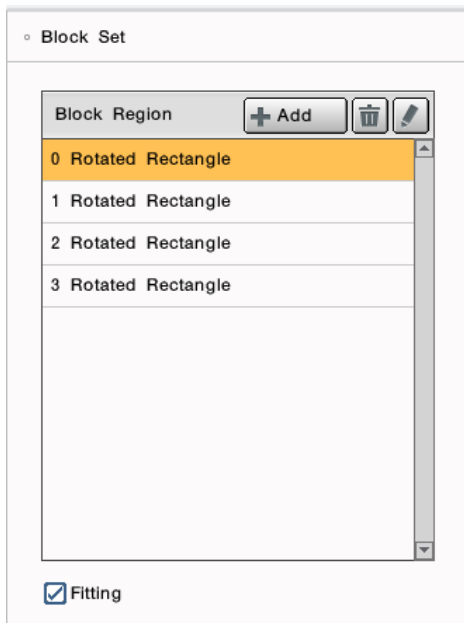
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.11.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

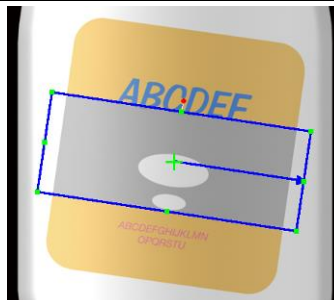
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.11.7. Step. 3 Library

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.11.8. Step. 4 Condition

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.11.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | 1622     |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

## 3.12. (CAM2) [T204]R1YS1Y



### 3.12.1. Summary

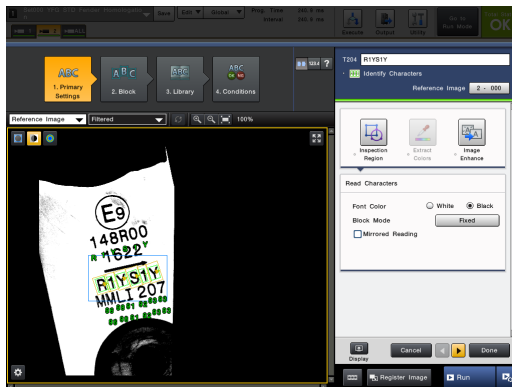
|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of this tool                                                                                                                                                   |
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

### 3.12.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.12.3. Tool Edit Screen View

This is the edit screen for the [T204]R1YS1Y tool.



#### [Step Buttons]

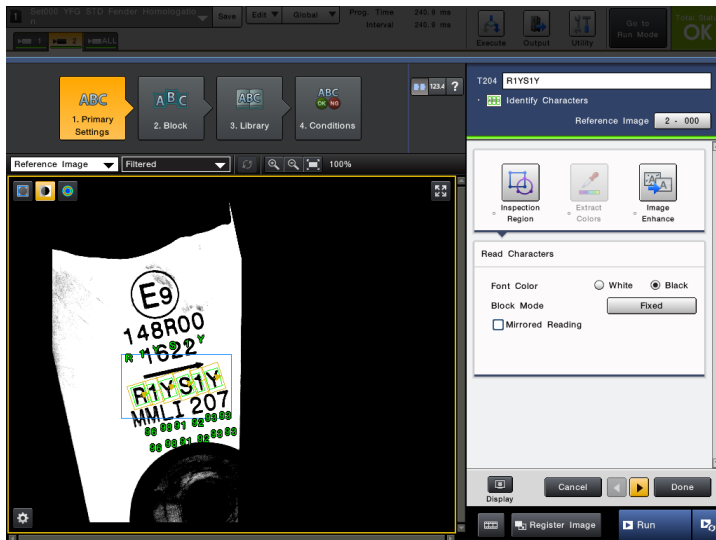
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                              |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                      | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.12.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                      | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.12.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                      | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.12.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                      | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.12.8 Step. 4 Condition</a> .                                         |                  |

### 3.12.4. Step. 1 Primary Settings

This is the edit screen for the [T204]R1YS1Y tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.12.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

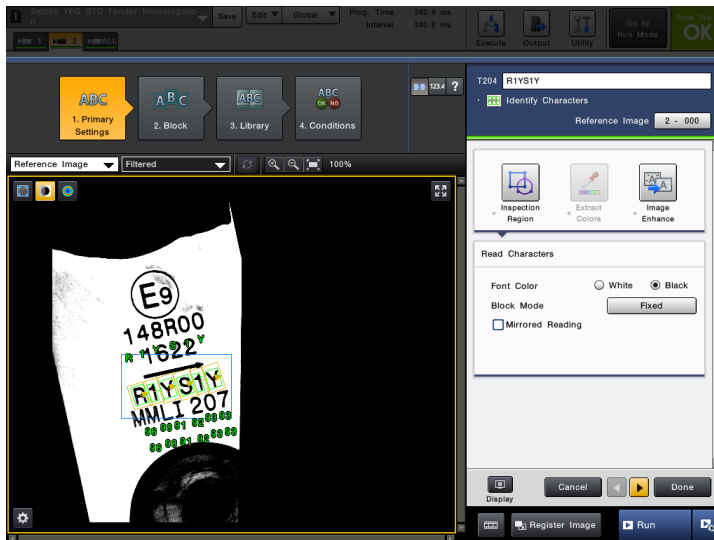
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.12.5. Setting the Inspection Region

#### 3.12.5.1. Inspection Region



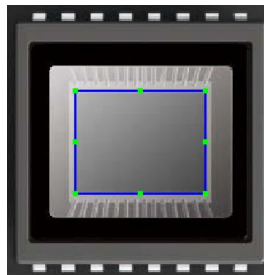
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

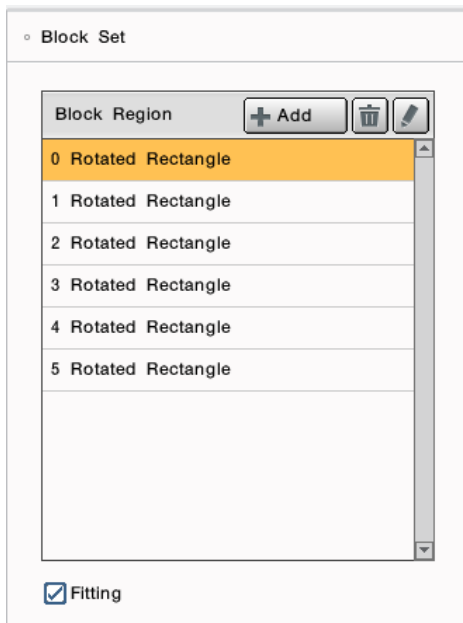
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.12.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

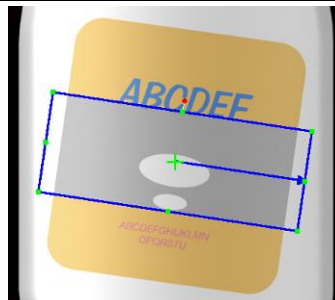
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.12.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library       |
|------------------------------|-------------------|
| Registered Characters        | 210 Character(s)  |
| Upper Case                   | 7 Type 80 Char.   |
| Lower Case                   | None              |
| Number                       | 9 Type 130 Char.  |
| Symbol                       | None              |
| (Built-in Library Characters | 166 Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.12.8. Step. 4 Condition

Judgment Conditions

Detected String  
R1YS1Y

Registered String

Correlation

|             |                                 |
|-------------|---------------------------------|
| Maximum     | 98                              |
| Minimum     | 91                              |
| Lower Limit | <input type="text" value="60"/> |

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.12.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | R1YS1Y   |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.13. (CAM2) [T205]MMLI207



#### 3.13.1. Summary

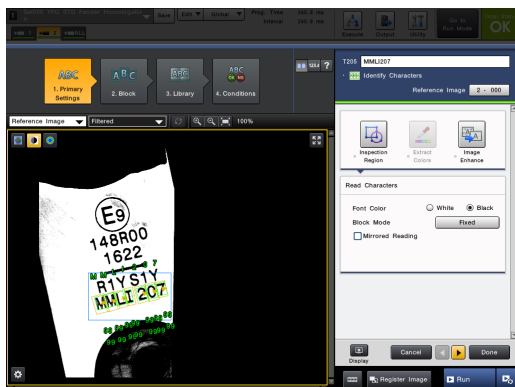
| Summary of this tool                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracts character information in the inspection region and matches them against the library data.<br>In this way, the string in the captured image can be identified. |
| Memo                                                                                                                                                                   |
|                                                                                                                                                                        |

#### 3.13.2. Defaults List of Parameters

| Parameter Name   | Defaults    | On Change | Memo |
|------------------|-------------|-----------|------|
| Block Mode       | Fixed       |           |      |
| Font Color       | Black       |           |      |
| Library Settings | Library No. |           |      |
| Library No.      | 0           |           |      |
| Mirrored Reading | OFF         |           |      |
| Fitting          | ON          |           |      |

### 3.13.3. Tool Edit Screen View

This is the edit screen for the [T205]MMLI207 tool.



#### [Step Buttons]

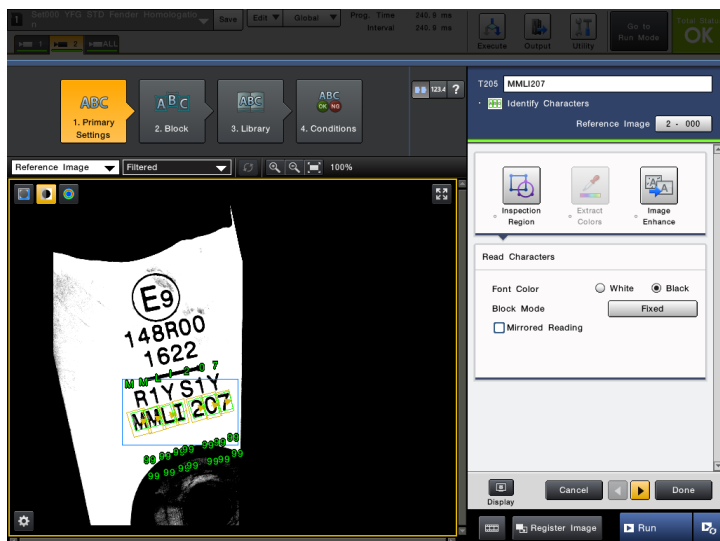
This tool is configured with the above Steps.

The Steps above are in the button mode. Clicking on the button changes the screen to the corresponding setting screen of each Step.

|                                                                                                                                                                                              |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Step. 1                                                                                                                                                                                      | Primary Settings |
| - Performs the settings regarding the Inspection Region, Extract Colors, Image Enhance and Read Characters.<br>Note: For details, refer to <a href="#">3.13.4 Step. 1 Primary Settings</a> . |                  |
| Step. 2                                                                                                                                                                                      | Block            |
| - Sets to block the character.<br>Note: For details, refer to <a href="#">3.13.6 Step. 2 Block</a> .                                                                                         |                  |
| Step. 3                                                                                                                                                                                      | Library Settings |
| - Registration to the library is made for the character pattern used in the recognize settings.<br>Note: For details, refer to <a href="#">3.13.7 Step. 3 Library</a> .                      |                  |
| Step. 4                                                                                                                                                                                      | Condition        |
| - Sets the judgment conditions with OK/NG-judgment to the recognized data.<br>Note: For details, refer to <a href="#">3.13.8 Step. 4 Condition</a> .                                         |                  |

### 3.13.4. Step. 1 Primary Settings

This is the edit screen for the [T205]MMLI207 tool (Step. 1).



#### [Inspection Region]

Sets the inspection region. For details, refer to [3.13.5 Setting the Inspection Region](#).



#### [Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool is/are as follows. Refer to [5.1 Image Enhance Settings](#) for the summary of each Image Enhance type.

#### Image Enhance List

0 Average 1 Blur 2 Contrast Expansion

#### [Read Characters]

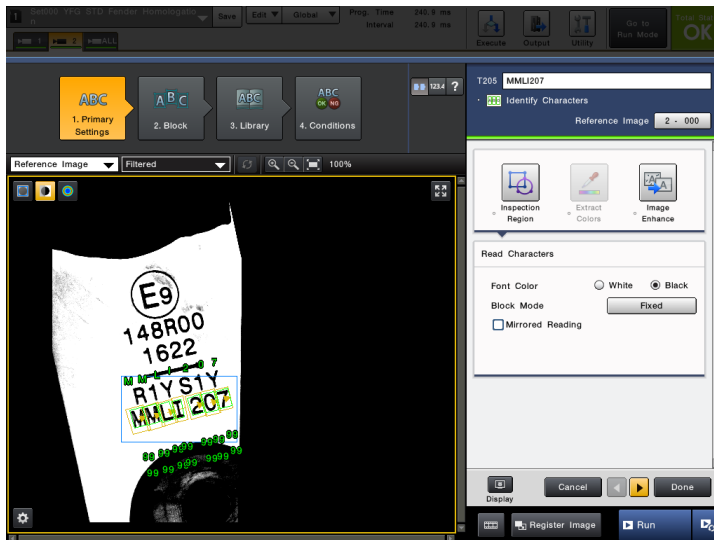
Font Color : Select the font color between [Black] and [White]

Block Mode : Sets the extraction mode that the character information in the inspection region is divided by one character.

Mirrored Reading : Check the box to support recognition of mirrored characters captured through the mirror or prism.

### 3.13.5. Setting the Inspection Region

#### 3.13.5.1. Inspection Region



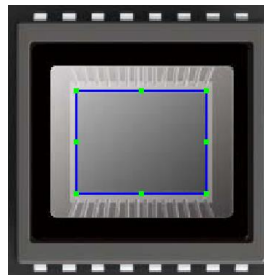
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

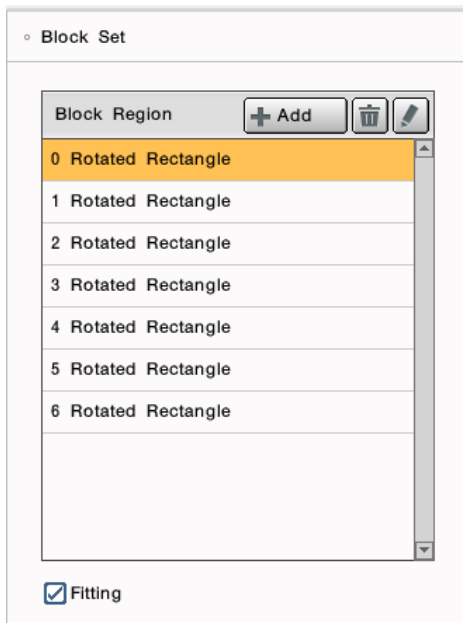
#### [Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.13.6. Step. 2 Block



Press the  button to add the block region. The region can be set every one letter according to the character.

To delete the region, select the relevant region and press the



button.

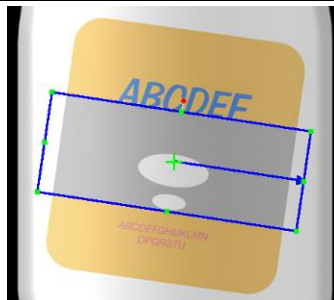
Refer to the following section for the extraction area drawing.

#### [Rotated Rectangle]

(How to setup)

Drag the red point to adjust the angle.

Then, drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



### 3.13.7. Step. 3 Library

Library Settings

☐ Built-in Library  
☒ Library No.

| Library Name                 | New Library      |
|------------------------------|------------------|
| Registered Characters        | 210 Character(s) |
| Upper Case                   | 7 Type 80 Char.  |
| Lower Case                   | None             |
| Number                       | 9 Type 130 Char. |
| Symbol                       | None             |
| (Built-in Library Characters | 166Character(s)) |

Selects the library to use for character recognition.

- Built-in Library: Uses the library that is built into the system.
- Library No.: Uses the library file of the specified number.

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

### 3.13.8. Step. 4 Condition

Judgment Conditions

Detected String  
MMLI207

Registered String

Correlation

Maximum 99

Minimum 99

Lower Limit

☐ Match With 1D/2D Code Read Data

Sets the Registered String to be used for judging the recognized contents and the tolerance for the recognition result measured value. If even a single character does not match between the recognition result and the Registered String, or the measured value is below the lower limit, the total status is judged as NG. If all the characters in the strings match and the measured value is equal to the lower limit or above, the total status is judged as OK.

#### 3.13.8.1. Defaults List of Judgment Conditions

| Parameter Name    | Defaults |
|-------------------|----------|
| Registered String | MMLI207  |
| Lower Correlation | 60       |

Note: For how to generally set the library, refer to [5.2 OCR2 Settings](#).

---

## 4. Position Adjustment

---

The position adjustment is the function that makes the inspection region automatically follows the object when it shifts in position. This is realized by specifying the reference tool so that the proper measurement can be performed even when the target object moves away from the initial position. It is also possible to specify individual reference tools for the positioning value of each individual component (X, Y,  $\theta$ ).

This setting uses the following tool(s) for the position adjustment.

| CAM 1 |                                             |                                                                    |
|-------|---------------------------------------------|--------------------------------------------------------------------|
| No.   | Reference Tool                              | Target Tool                                                        |
| 0     | T100 Pos. Adj. with Pattern Match (Profile) | T101 E9<br>T102 148R00<br>T103 1622<br>T104 R1YS1Y<br>T105 MMLI207 |

| CAM 2 |                                             |                                                                    |
|-------|---------------------------------------------|--------------------------------------------------------------------|
| No.   | Reference Tool                              | Target Tool                                                        |
| 0     | T200 Pos. Adj. with Pattern Match (Profile) | T201 E9<br>T202 148R00<br>T203 1622<br>T204 R1YS1Y<br>T205 MMLI207 |

---

## 5. Supplement

---

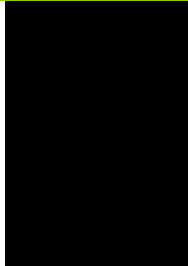
### 5.1. Image Enhance Settings

---

#### [Average]

##### (General Overview)

“Average” is a filter which has an effect to make an image smooth. This filter is used for raising the precision of the edge detection without being subject to noise influence.



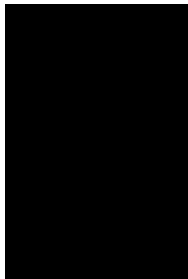
Noises are captured along the object's end.



##### (Usage)

“Count” sets the strength of the filter effect. Heighten the filter effect to make an image smoother.

Increase “Count” value, and the filter effect becomes higher.



The object's end turns smoother.

---

## [Blur]

### (General Overview)

“Blur” is a filter which has an effect to blur images, and is used for smoothing images much more than “Average” filter.



The object's surface looks very white grainy.

### (Usage)

Set the strength of the filter effect by setting “Intensity”. Heighten the filter effect, and a more blurred image can be made.

The more “Intensity” is increased, the more the filter effect is strengthened.



The surface turns smoother.

## [Contrast Expansion]

(General Overview)

Expands the distribution of brightness in the image to increase the contrast.

(Usage)

“Expansion Width”: Adjusts the intensity of the contrast expansion.

“Noise Cut”: Reduces noise. Noise occurs particularly in the dark parts of images. When this value is increased, the occurrence of noise can be suppressed more easily but it will also be easier for the image to become blurred.



Entire image is dark with poor visibility.

↓

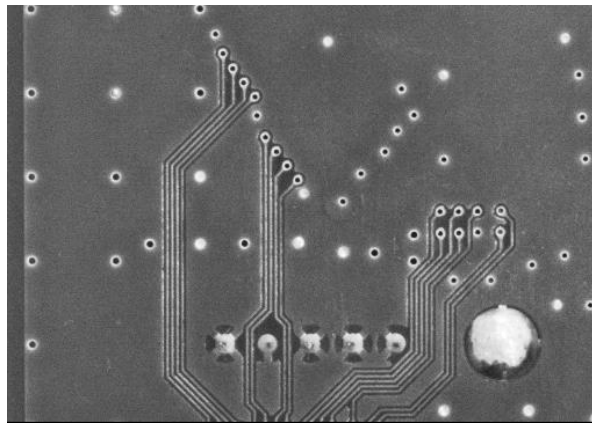


Image is improved to have a good balance of brightness and good visibility.

## 5.2. OCR2 Settings

[Library Settings]

## <General Overview>

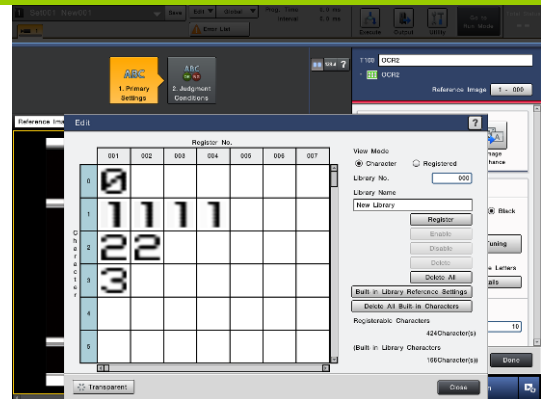
The library is the collection of character patterns which are used for recognizing characters. Register one or more characters for the recognition based on an image.

<Basic procedure>

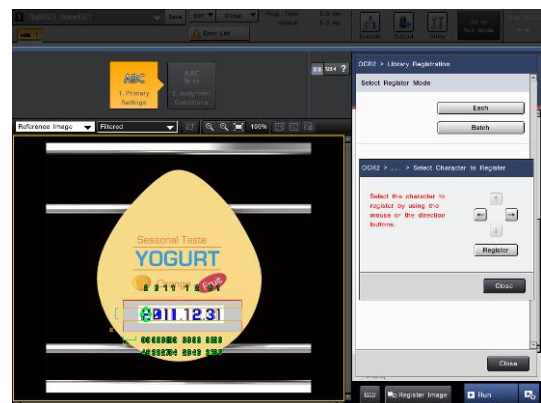
1. Left-click the “Edit” button and set the library of the selected library No.
  2. Left-click the “Register” button. A character pattern can be newly added to the library.
  3. “Each”: Specifies and adds one of the recognized character patterns.
- “Batch”: Adds all recognized character patterns at one time.

4. When selecting “Each”, specify the character pattern that is to be added and add it to the library.

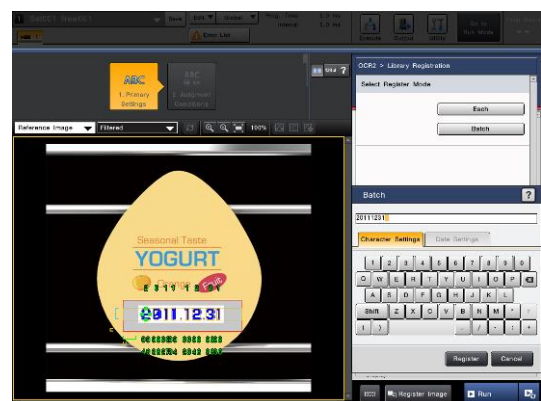
When selecting “Batch”, enter the recognized character patterns in order and add them to the library.



[OCR Library] is a collection of character patterns.



Using "Each" specifies and registers the recognized character patterns.



Using "Batch" registers all the recognized character patterns.

## [Judgment Conditions]

### <General Overview>

“Judgment Conditions” is where the text string to be judged as OK is set.

Mainly, the following items can be used as judgment conditions:

1. Fixed text using alphanumeric characters
2. Date based on the built-in clock
3. Alphanumeric characters with date encrypted

### <Basic procedure>

1. To use fixed text as judgment conditions...

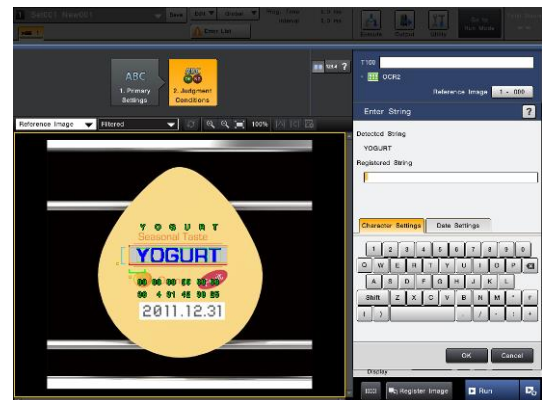
Click the text entry box to enter the text through the keyboard.

To use the recognized text string itself as the judgment condition...

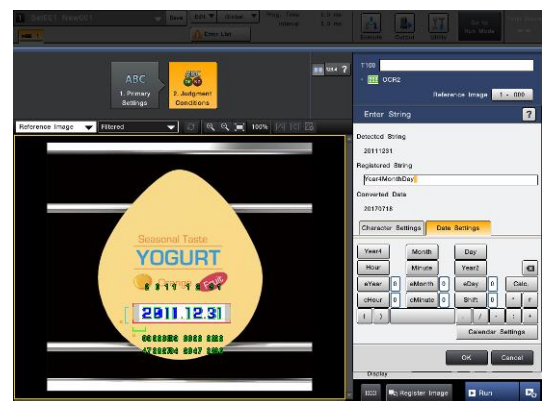
Press “Copy Detected String” button, which is provided for convenience.

2. To use the date based on the built-in clock as judgment conditions...

Click the text entry box and select “Special Character”.  
“Year4”, “Month”, “Day”, “Hour”, “Minute” and “Year2” are the year (the four digits of the year), month, day, hour, minute and year (the last two digits of the year) of the inspection time, respectively.



The alphabet and numeric characters can be entered for the judgment conditions.



Date at the time of inspection can be entered for the judgment conditions.

## [Judgment Conditions]

3. To use alphanumeric characters with date encrypted as judgment conditions...

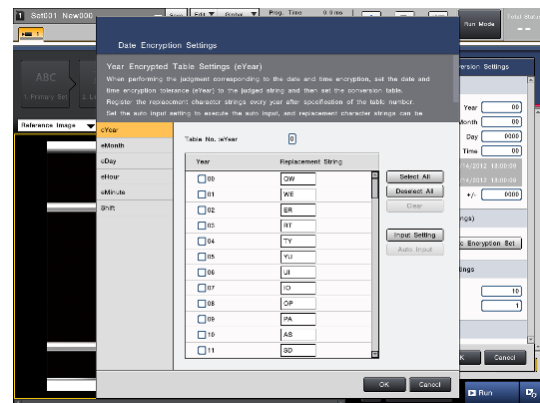
Click the text entry box and select “Special Character”.

“eYear”, “eMonth”, “eDay”, “eHour” and “eMinute” enters the year (the last two digits of the year), month, day, hour and minute of the current inspection time, respectively. In addition, “Shift” enters the fixed time zone (specified by minute). Combination with fixed text is possible.

Use “Date Encryption Settings” under “Calendar Settings” to set the encryption corresponding to the year, month, day, hour and minute using alphanumeric characters of four digits or less.



The date-encrypted alphanumeric characters can be entered for the judgment conditions.



“Date Encryption Settings” is used for correspondence between the date and alphanumeric characters.

---

## 6. Appendix

---

### 6.1. Program Export/Import

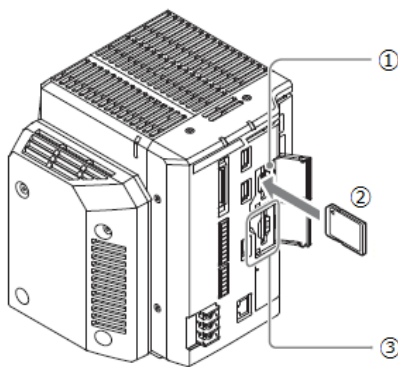
This appendix serves as a guide on exporting program setting files and image files to the SD Card.

By following the operation procedure in this appendix, files needed for simulation in remote locations are exported to the SD Card.

#### 6.1.1. Preparation

##### Inserting an SD Card

Insert the SD card into the SD1 and SD2 slot such that the triangle inscription is on the top of the card.



1. SD2 slot
2. Insert SD card.
3. SD1 slot

(Do not insert or remove while power is on.)

##### Supported SD Card Models

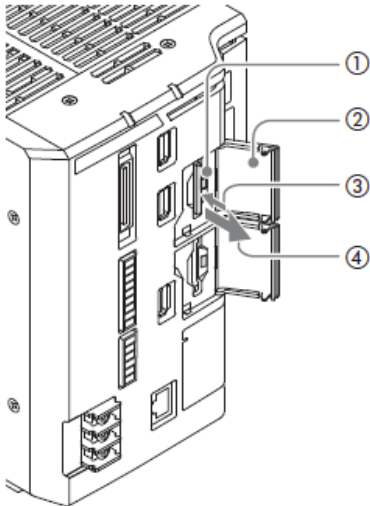
Use the Keyence optional OP-87133 (512 MB), optional CA-SD1G (1 GB), optional CA-SD4G (4 GB), or optional CA-SD16G (16 GB). Note that the controller ships with an OP-87133 (CA-SD1G for CV-X350/X450/X470/X480/X480LJ and CA-SD4G for CV-X490) inserted in the SD1 slot.

Note: SD cards other than the models mentioned above may not operate normally and are exempt from the warranty of this equipment.

## Removing an SD Card



Choose [Utility] > [Remove External Media] > [Remove SD Card 2] to turn off the power of SD Card 2. Then, press the SD card in the SD2 slot inward to release and remove the card.



1. Make sure the access lamp is no longer lit.
2. Open cover.
3. Press the SD card to unlock it.
4. Remove SD card.

## 6.1.2. Archived Image

### What is [Archived Image]

An archived image is an image temporarily saved in the internal memory of the CV-X. Images are archived and accumulated during operation.

When performing [Exporting the Current Program with Archived Images] and [Exporting Archived Images Only], it is necessary that there are images archived. Verify the images using the procedure below.

### Accumulating Archived Images

Images are automatically archived whenever a trigger is input during Run Mode.

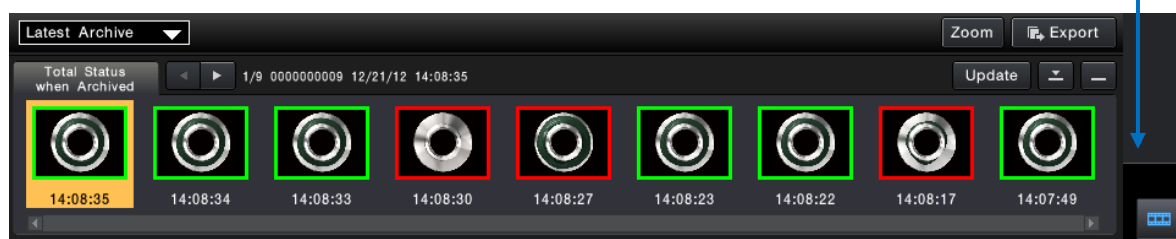


Click the button on the upper right portion of the screen to switch to Run Mode and input a trigger.

### Verifying Archived Images

Click the button on the lower right portion of the screen to open the Image Strip.

If there were images archived, they will be displayed as thumbnails.



### 6.1.3. Export

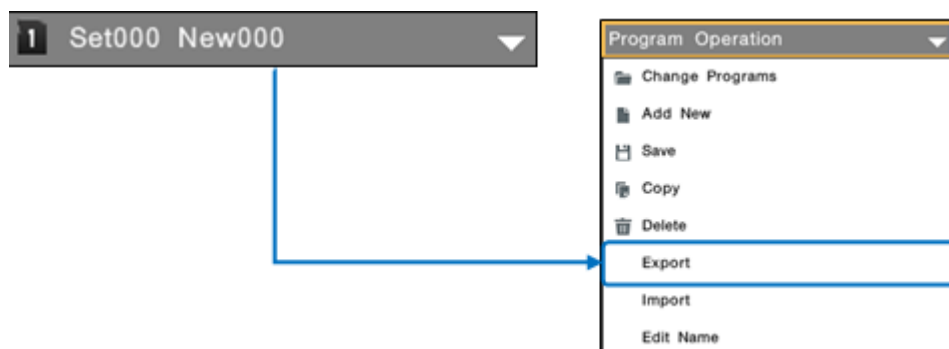
#### Exporting a Program Setting

This is the procedure for exporting a program setting to the SD card.



Click the button on the upper right portion of the screen to switch to Setup Mode.

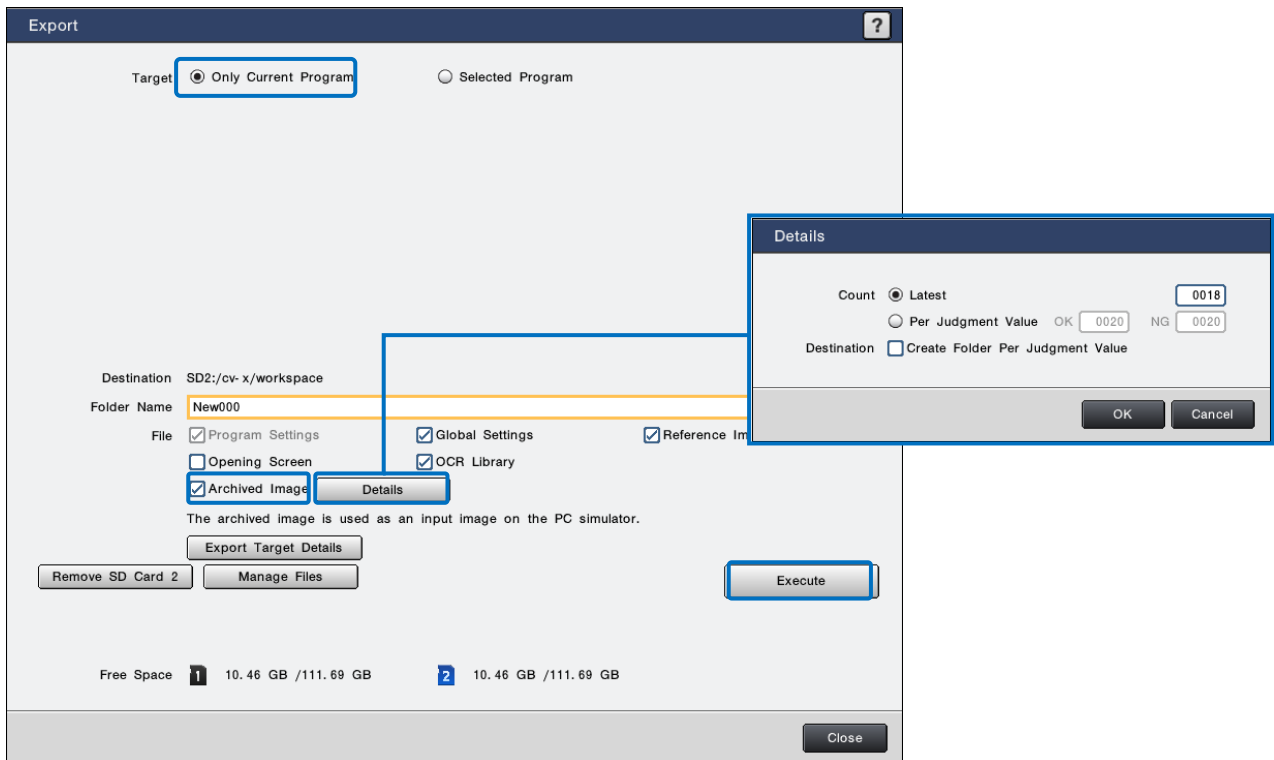
Click the drop-down menu on the upper left of the screen and choose [Export] from the Program Operation menu that appears.



#### (1) In the Case of Exporting the Current Program with Archived Images

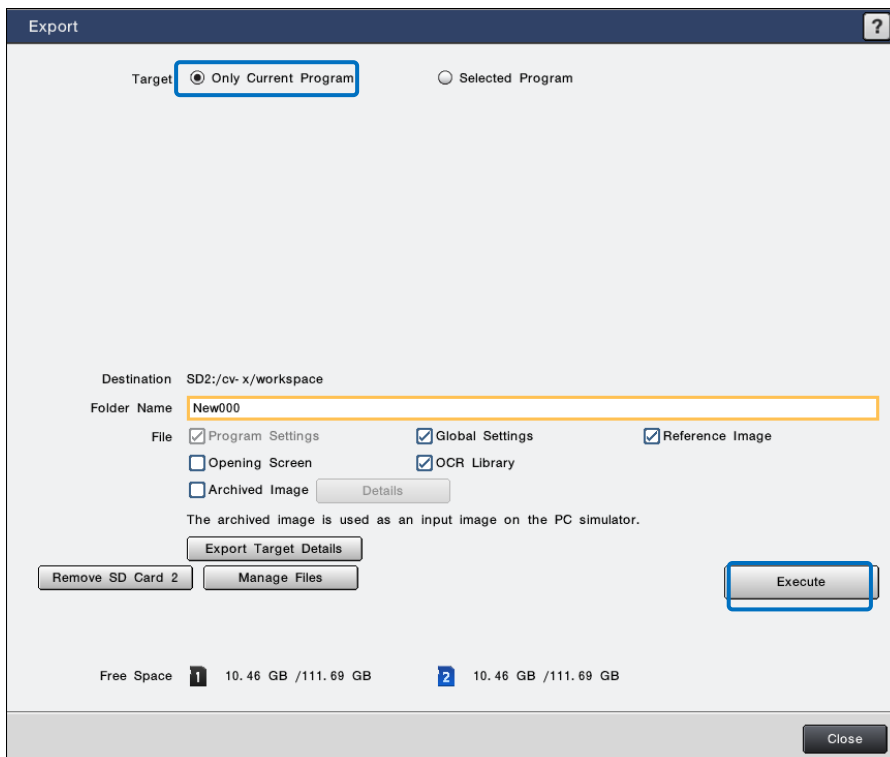
Select [Only Current Program]. Check [Archived Image] under [File], left-click , and specify the number of images to export. Then, click the [Execute] button.

Note: The latest archived images of the specified number are exported together with the program setting. Decide on the number of images to export while paying attention to the remaining capacity of the SD card.



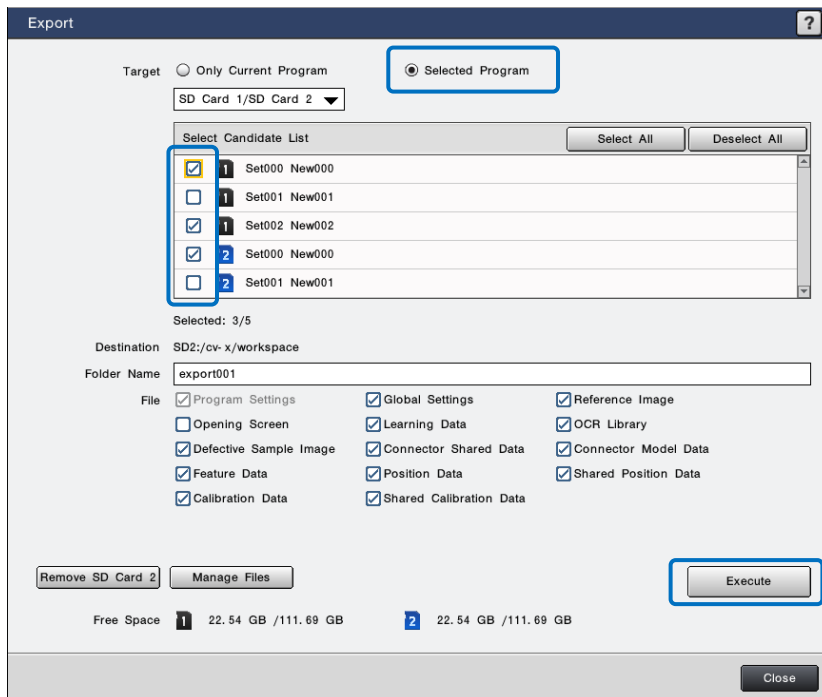
## (2) In the Case of Exporting the Current Program Only

Select [Only Current Program] and click on the [Execute] button. Choose the target files to export as necessary.



### (3) In the Case of Exporting Selected Programs

Select [Selected Program]. Put a check mark on the desired program settings to export and then click the [Execute] button. Choose the target files to export as necessary.



#### Details of Export Target [File]

| File                   | Explanation                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Program Settings       | The program setting file (inspect.dat) of tools etc.                                                                                                |
| Global Settings        | The global settings data (env.dat) such as Input/Output etc.                                                                                        |
| Reference Image        | The reference images (ref*_*.bmp [.jpg] [.png]).                                                                                                    |
| Opening Screen         | The display image data (Logo.bmp) shown at start-up.                                                                                                |
| OCR Library            | The library data (dic*.dat) used in the OCR tool.                                                                                                   |
| Archived Image         | Exports the specified number of latest or specified type of archived images (Available only when [Only Current Program] is selected as the Target). |
| Learning Data          | The learning data (SCT.tbd) of Auto-Teach Inspection.                                                                                               |
| Defective Sample Image | The images (No_NGMASTER_CAM No_NG.bmp [.jpg] [.png]) set as defective samples in Auto-Teach Inspection.                                             |

|                          |                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Feature Data             | The feature data (STF.tbd) set in Feature Drawing Tool.                                                   |
| Shape Region Data        | The shape region data (RGN_SHP.dat) set in Shape Region.                                                  |
| Outline Setting Data     | The outline setting data (CNT_SHP.dat) set in Outline Comparison.                                         |
| Subtraction Setting Data | The subtraction setting data (TDS_MDL.dat) set in 3D Comparison.                                          |
| Camera Calibration Data  | The calibration data (TDC_G_CLB_*.dat) used in 3D Capture Mode.                                           |
| Position Data            | The position data (RBT_L_POS_*.dat) used in Vision-Guided Robotics Setting.                               |
| Shared Position Data     | The shared position data (RBT_G_POS_*.dat) used in Vision-Guided Robotics Setting.                        |
| Calibration Data         | The calibration data (RBT_L_CLB_*.dat, 3D_RBT_L_CLB_*.dat) used in Vision-Guided Robotics Setting.        |
| Shared Calibration Data  | The shared calibration data (RBT_G_CLB_*.dat, 3D_RBT_G_CLB_*.dat) used in Vision-Guided Robotics Setting. |
| Layout Data              | The layout data (LYT_G_*.tbd) used in 3D Vision-Guided Robotics Setting.                                  |
| Robot Model Data         | The robot model data (RBT_G_RMD_*.dat) used in 3D Vision-Guided Robotics Setting.                         |
| Connector Model Data     | The model data (mdl_l_*.tbd) registered in the Connector Dimension Inspection tool.                       |
| Connector Shared Data    | The data (mdl_s_*.tbd) registered in the Share Settings of the Connector Dimension Inspection tool.       |

## Exporting Archived Images Only

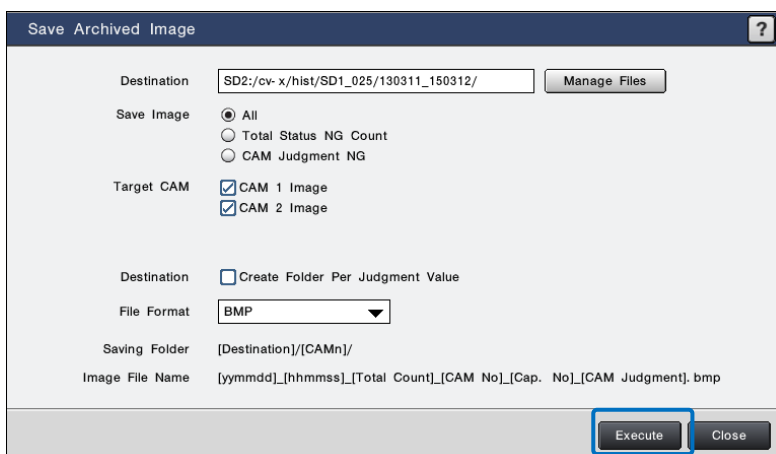
Click the button  on the lower right of the screen to open the Image Strip. Then, click on the [Export] button to make the images selectable.



Check the desired images for export (Pressing the [Select All] button puts a check mark on all the images). Upon clicking the [Execute] button, the [Save Archived Image] dialog appears.



Make the appropriate selection for [Save Image], [Target CAM], [Destination] and [File Format] and then click on the [Execute] button.



Save Image (This can be set when [Select All] is chosen in the Image Strip)

| Option                       | Explanation                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>All</b>                   | Exports all the images.                                                                                 |
| <b>Total Status NG Count</b> | Exports the images of all cameras of measurement counts where the Total Status was NG during archiving. |
| <b>CAM Judgment NG</b>       | Exports only the images where CAM Judgment was NG during archiving.                                     |

## 6.1.4. Import

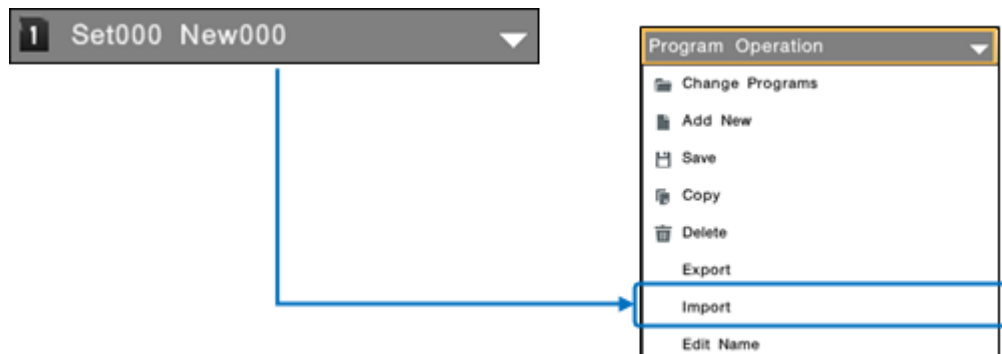
### Importing a Program Setting

This is the procedure for importing a program setting from the SD card.



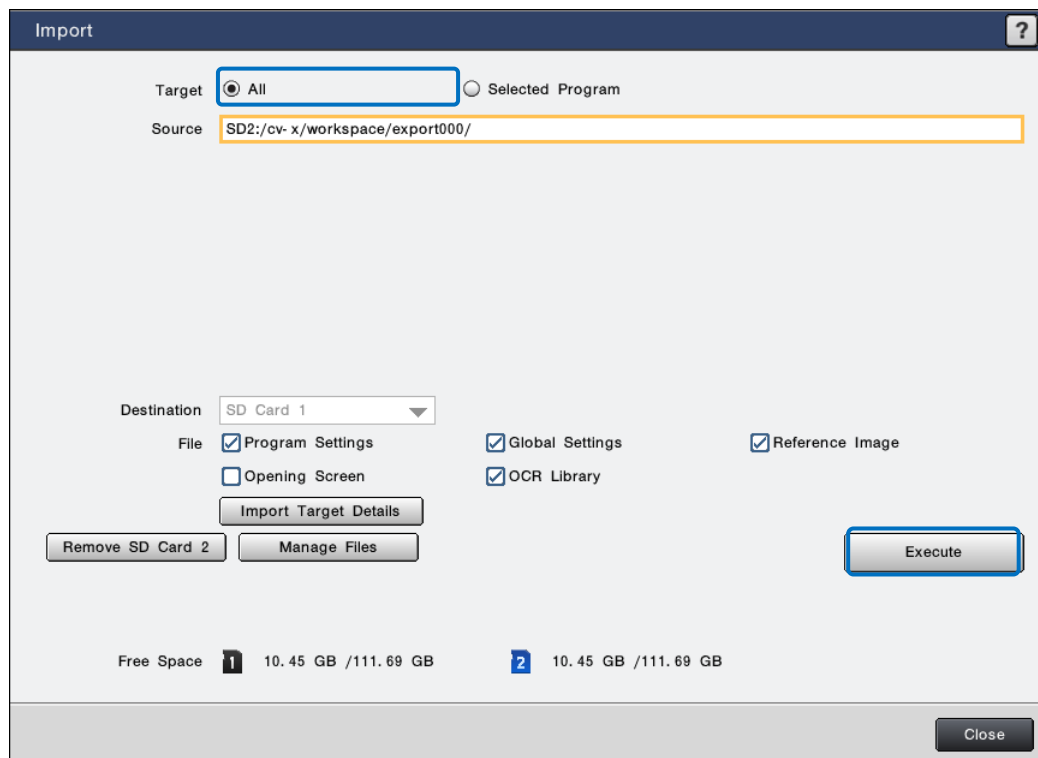
Click the button on the upper right portion of the screen to switch to Setup Mode.

Click the button on the upper left of the screen and choose [Import] from the Program Operation menu that appears.



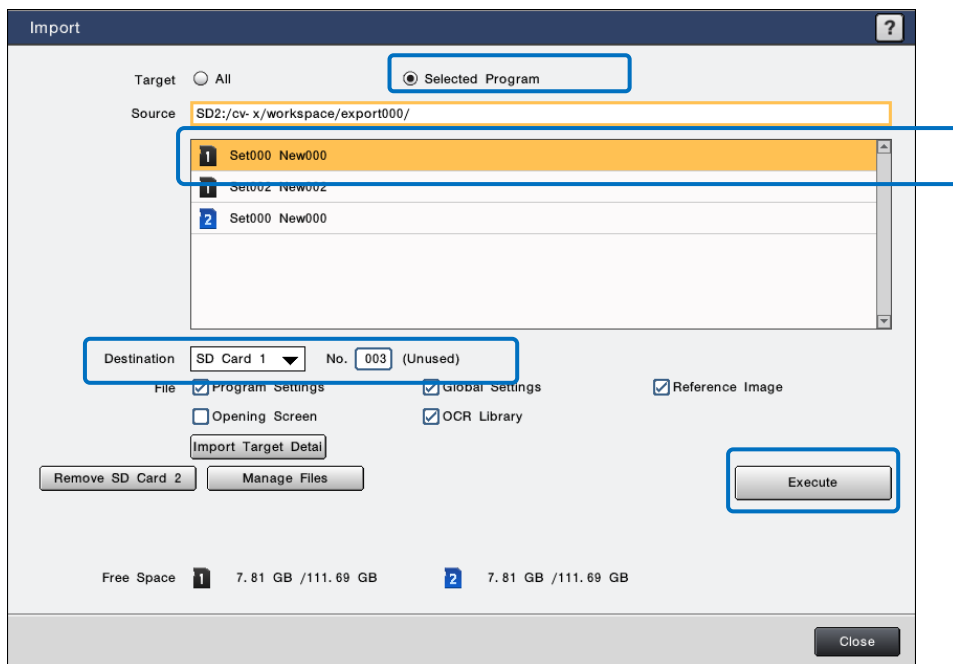
#### (1) In the Case of Importing All Programs Within the Workspace

Select [All]. Specify the target workspace folder in [Source] and then click the [Execute] button. Choose the target files to import as necessary.



## (2) In the Case of Importing a Selected Program from the Workspace

Select [Selected Program] and specify the target workspace folder in [Source]. Choose the desired program setting to import from the list, specify the [Destination] to put it to, and then click the [Execute] button. Choose the target files to import as necessary.



### Details of Import Target [File]

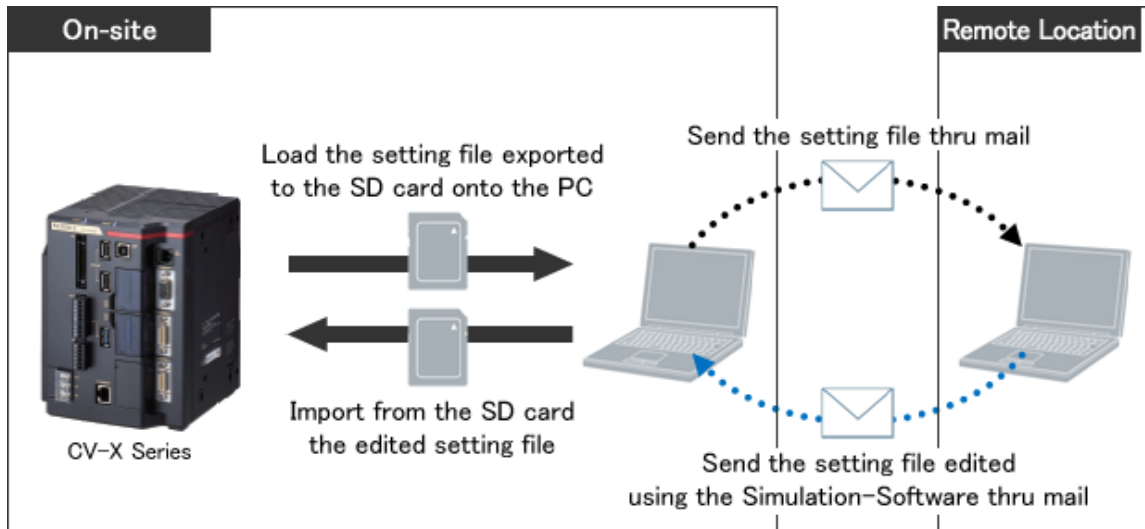
| File                   | Explanation                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Program Settings       | The program setting file (inspect.dat) of tools etc.                                                    |
| Global Settings        | The global settings data (env.dat) such as Input/Output etc.                                            |
| Reference Image        | The reference images (ref*_*.bmp [.jpg] [.png]).                                                        |
| Opening Screen         | The display image data (Logo.bmp) shown at start-up.                                                    |
| OCR Library            | The library data (dic*.dat) used in the OCR tool.                                                       |
| Learning Data          | The learning data (SCT.tbd) of Auto-Teach Inspection.                                                   |
| Defective Sample Image | The images (No_NGMASTER_CAM No_NG.bmp [.jpg] [.png]) set as defective samples in Auto-Teach Inspection. |
| Feature Data           | The feature data (STF.tbd) set in Feature Drawing Tool.                                                 |

|                          |                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Shape Region Data        | The shape region data (RGN_SHP.dat) set in Shape Region.                                                  |
| Outline Setting Data     | The outline setting data (CNT_SHP.dat) set in Outline Comparison.                                         |
| Subtraction Setting Data | The subtraction setting data (TDS_MDL.dat) set in 3D Comparison.                                          |
| Camera Calibration Data  | The calibration data (TDC_G_CLB_*.dat) used in 3D Capture Mode.                                           |
| Position Data            | The position data (RBT_L_POS_*.dat) used in Vision-Guided Robotics Setting.                               |
| Shared Position Data     | The shared position data (RBT_G_POS_*.dat) used in Vision-Guided Robotics Setting.                        |
| Calibration Data         | The calibration data (RBT_L_CLB_*.dat, 3D_RBT_L_CLB_*.dat) used in Vision-Guided Robotics Setting.        |
| Shared Calibration Data  | The shared calibration data (RBT_G_CLB_*.dat, 3D_RBT_G_CLB_*.dat) used in Vision-Guided Robotics Setting. |
| Layout Data              | The layout data (LYT_G_*.tbd) used in 3D Vision-Guided Robotics Setting.                                  |
| Robot Model Data         | The robot model data (RBT_G_RMD_*.dat) used in 3D Vision-Guided Robotics Setting.                         |
| Connector Model Data     | The model data (mdl_l_*.tbd) registered in the Connector Dimension Inspection tool.                       |
| Connector Shared Data    | The data (mdl_s_*.tbd) registered in the Share Settings of the Connector Dimension Inspection tool.       |

## 6.1.5. Exchange of Setting File

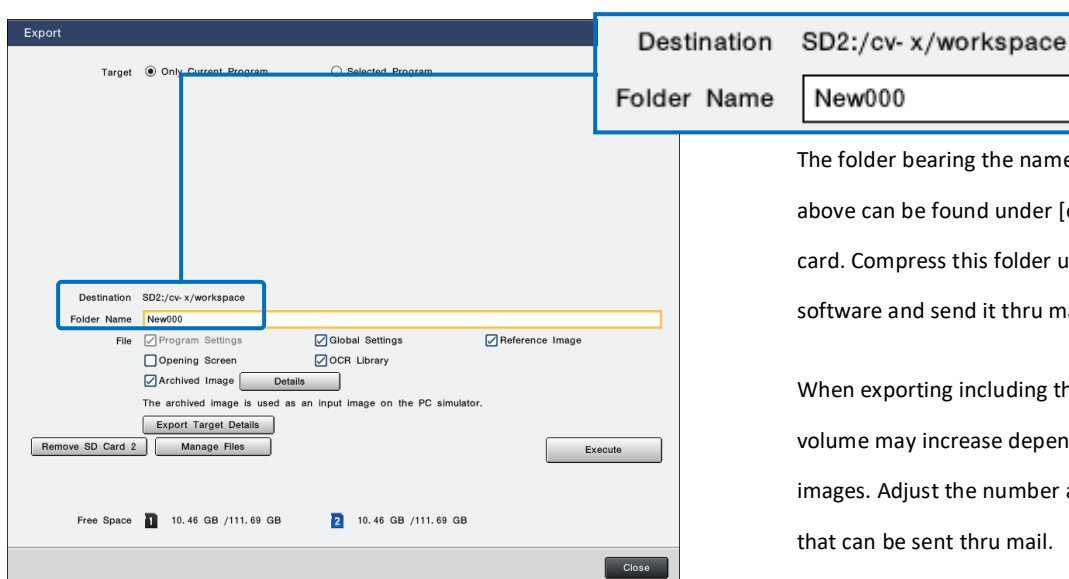
### Sending the Setting File thru Mail

The exported setting file can be loaded into the [CV-X Series Simulation-Software] and can be edited and saved just as in a controller. For example, when product type addition or change of settings need to be made and the creator of the setting is in a remote location, the problem can be solved in the shortest possible time by exchange of the setting file thru mail.



### Where the File (Exported Setting File) to Send thru Mail Is

The exported setting file is saved in the SD card in a folder structure called [Workspace] which can be loaded onto the [CV-X Series Simulation-Software] as is. When exchanging the setting file thru mail, it is convenient to send the [Workspace]. For example, when exporting a setting using the procedure described in [(1) In the Case of Exporting the Current Program with Archived Images] under section [3. Export]



The folder bearing the name specified in [Folder Name] above can be found under [cv-x]>[workspace] of the SD card. Compress this folder using a compression software and send it thru mail.

When exporting including the archived images, the volume may increase depending on the number of images. Adjust the number according to the amount that can be sent thru mail.

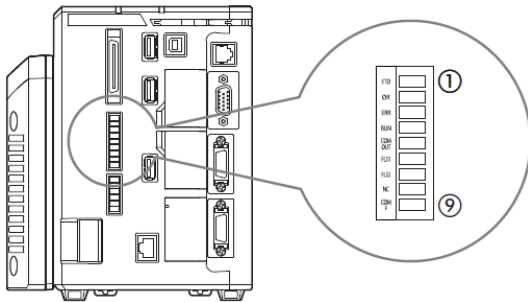
## 6.2. Outputting Total Status

### 6.2.1. Performing Wiring

When outputting the Total Status of the inspection, perform the wiring below.

Here, the wiring of the terminal blocks [OUT22: STO] and [OUT23: OR] to a PLC or a relay will be explained.

#### Controller Terminal Block (OUT Connector)



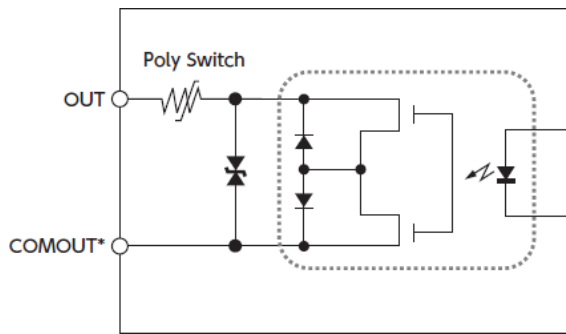
| No. | Terminal Name *1 |
|-----|------------------|
| 1   | OUT22 (STO)      |
| 2   | OUT23 (OR)       |
| 3   | F_OUT2 (ERR)     |
| 4   | F_OUT3 (RUN)     |
| 5   | COMOUT1 (COMOUT) |
| 6   | F_OUT0 (FLS1)    |
| 7   | F_OUT1 (FLS2)    |
| 8   | N.C.             |
| 9   | COMOUT_F (COM1)  |

Suitable wiring: AWG16 to 28

Terminal block screw torque: 0.25Nm or less

\*1 The words in () are the words printed on the accompanying terminal blocks during shipment.

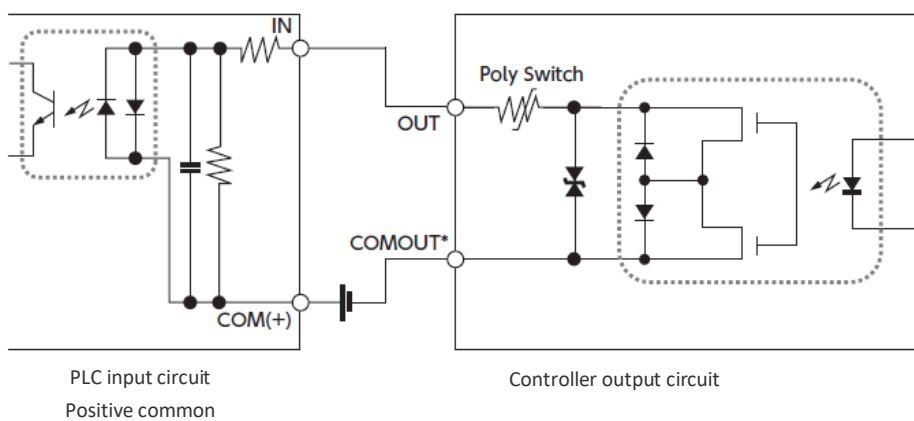
## Output Circuit Diagram



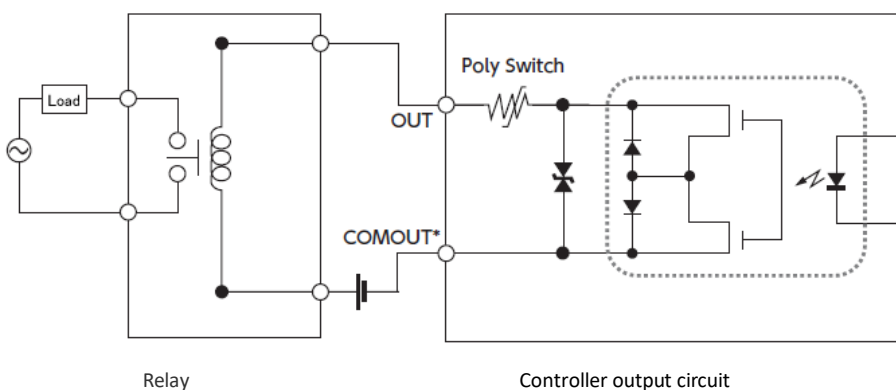
- Maximum applied voltage: 30V
- Maximum sink current: 50mA
- Leakage current: 0.1mA or less
- Residual voltage:
  - 1.4V or less (50mA)
  - 1.0V or less (20mA)

## Examples of Connections

When connecting an output from the controller to a PLC with a positive common



When connecting an NPN output from the controller to a relay



## 6.2.2. Confirming External Terminal Connection Status (I/O Monitor)

The connection status of signals inputted/outputted via the CV-X400 /X300Series External Terminal can be confirmed.

Click the button on the upper right-hand side of the screen to switch to Setup Mode.

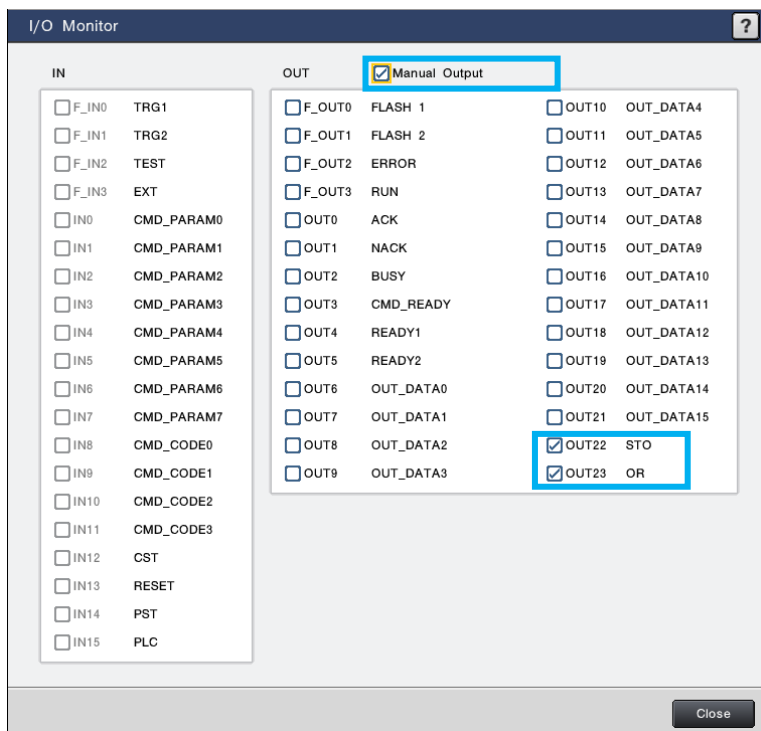


Select [I/O Monitor] from [Utility] to open the I/O Monitor.



The desired output terminals can forcibly be turned on and signal confirmation can be done on the connected device by putting a checkmark on [Manual Output]. Forced output can only be used in Setup Mode.

Put a checkmark on [OUT22 (STO)] and [OUT23 (OR)], and confirm whether the ON/OFF statuses of the signals can be recognized by the connected device (PLC, relay, etc.). Putting a checkmark turns the status to ON (Short) and removing the checkmark turns the status to OFF (Open).



When the ON/OFF status linked to checking/unchecking cannot be confirmed, check the wiring once again.

### 6.2.3. Setting the Tools to Include in Total Status Judgment

The tools to include in Total Status judgment can be arbitrarily selected. This is useful when performing of Total Status judgment using only the judgment results of particular tools is desired.

Click the button on the upper right-hand side of the screen to switch to Setup Mode.

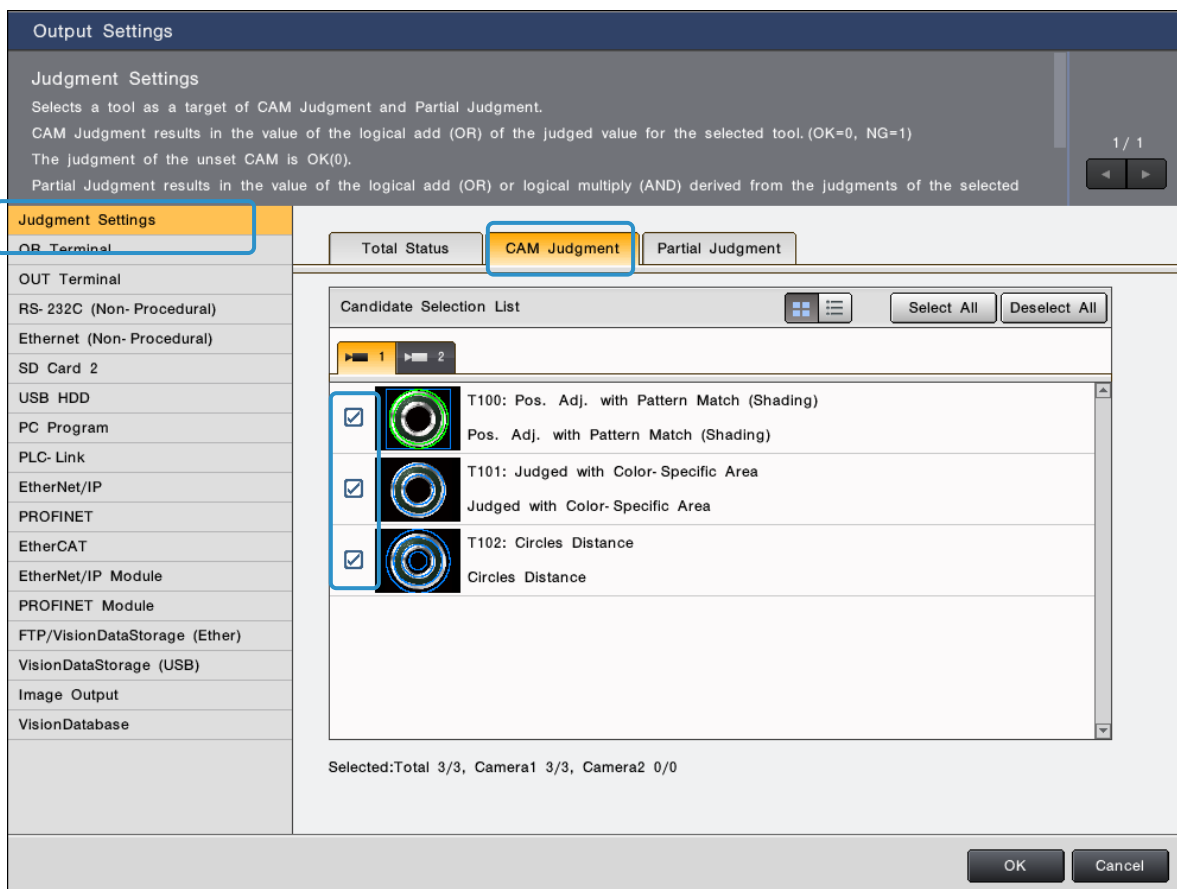


Open [Output].



Select [Judgment Settings] > [CAM Judgment].

In [Select Candidate List], verify that the tools to include in Total Status judgment are checked. (When using multiple cameras, switch between the camera tabs and perform the same verification.)



## 6.2.4. Setting OR Terminal Output Mode

When outputting the Total Status via the OR Terminal, select the terminal output mode.

Select [OR Terminal], then set the [OR Terminal Output Mode] and [OR Terminal Output Timing].

The screenshot shows the 'Output Settings' dialog box. The 'OR Terminal Output Settings' section is active, displaying instructions for the terminal output state. Below this, a list of 'Judgment Settings' includes 'OR Terminal' (highlighted), 'OUT Terminal', and various communication methods like RS-232C, Ethernet, SD Card, USB HDD, PC Program, PLC-Link, EtherNet/IP, PROFINET, EtherCAT, EtherNet/IP Module, PROFINET Module, FTP/VisionDataStorage (Ether), VisionDataStorage (USB), Image Output, and VisionDatabase. The 'OR Terminal Output Mode' section has three radio buttons: 'Update Every Judge' (selected), 'Hold Status', and 'One Shot Output'. The 'OR Terminal Output Timing' section has two radio buttons: 'Output on NG Status' (selected) and 'Output on OK Status'. A 'One Shot Time (ms)' input field is set to '0500'. At the bottom right are 'OK' and 'Cancel' buttons.

### OR Terminal Output Mode

- **Update Every Judge:** The OR terminal output state varies in response to the judgment result at every judgment finalization. Reading is synchronized to the rise of the STO outputted at every judgment finalization.

- **Hold Status:** Once an NG status is output, the NG status is maintained regardless of the judgment results that come after.

- **One Shot Output:** The OR terminal turns on only for the period set in [One Shot Time] and then it turns off after the elapse of the set time.

### OR Terminal Output Timing

- **Output on NG Status**

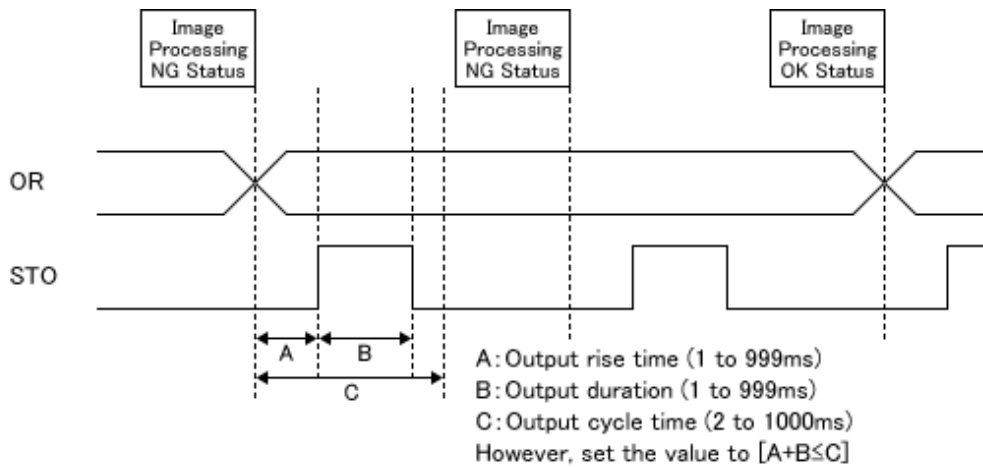
- **Output on OK Status**

Refer to the next page and following for the timing chart of each output mode.

## 6.2.5. Timing Chart

The timing chart for each OR terminal output mode can be seen below.

### · Update Every Judge



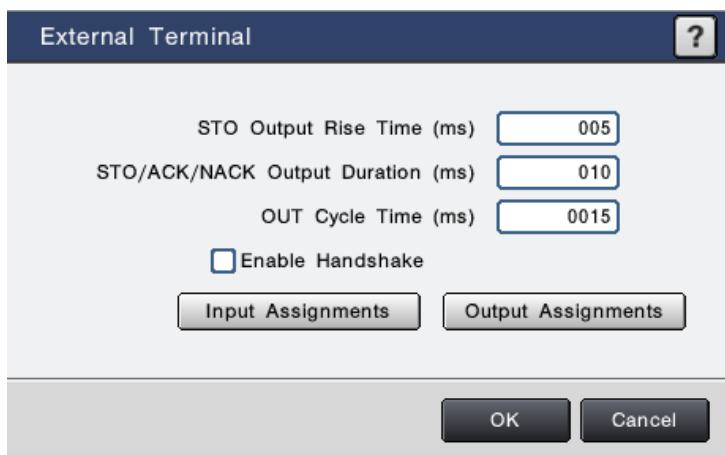
Since STO (OUT22) rises whenever a judgment is finalized, check the ON/OFF status of OR (OUT23) in synchronization to it. The STO output time (A/B/C in the timing chart) can be changed in the global settings. Adjust the time to a time that can be recognized by the connected PLC.

### How to Change STO Output Time

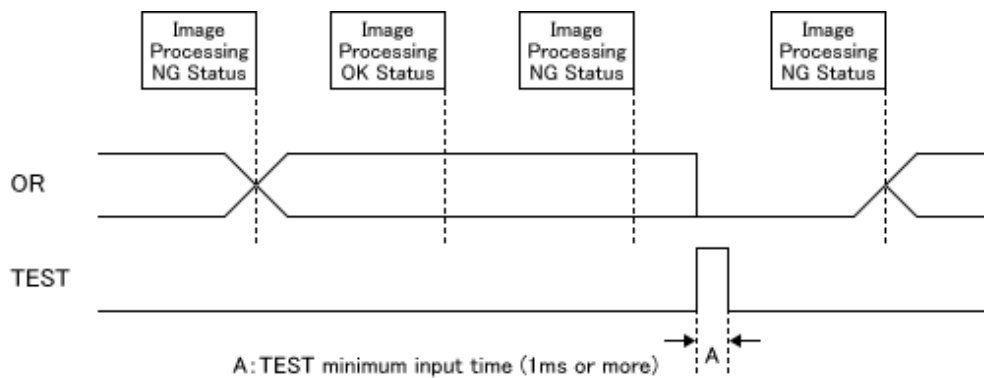
Select [Global] > [Communications & I/O] > [External Terminal].



Adjust A: Output rise time (1 to 999ms), B: Output duration (1 to 999ms) and C: Output cycle time (2 to 1000ms) to a time that can be recognized by the connected PLC. However, set the value to  $A+B \leq C$ .



#### · Hold Status

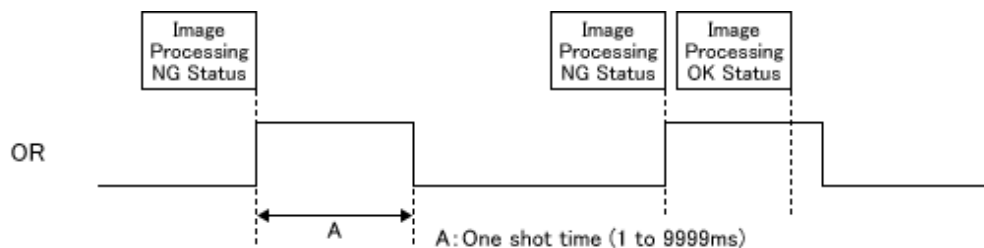


In normal circumstances, judgment output status is updated upon every trigger. But in this mode, after an NG Status occurs, it is held until it is released by TEST input etc.

Once an NG occurs, the NG status of the OR output does not change regardless of the actual judgment result.

When the TEST input is turned on/off, the hold status is released and all terminals are reset to the normal status. The OR output status will be updated when the result for the next OR output is output.

#### · One Shot Output



The OR output holds the NG status for the duration of the set time, and then releases the NG status afterwards. When continuous image processing is performed in a state where One Shot Output is set, there is a chance that the previous NG status is still held during the judgment result finalization time of STO-based image processing.

STD FENDER HOMOLOGATION

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